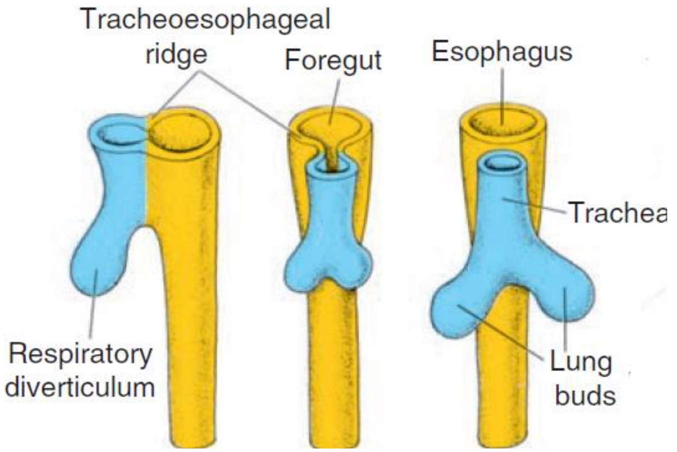
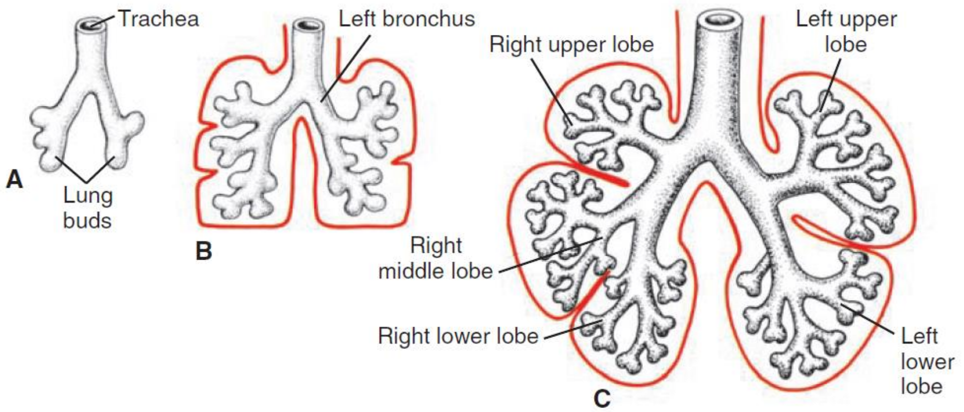
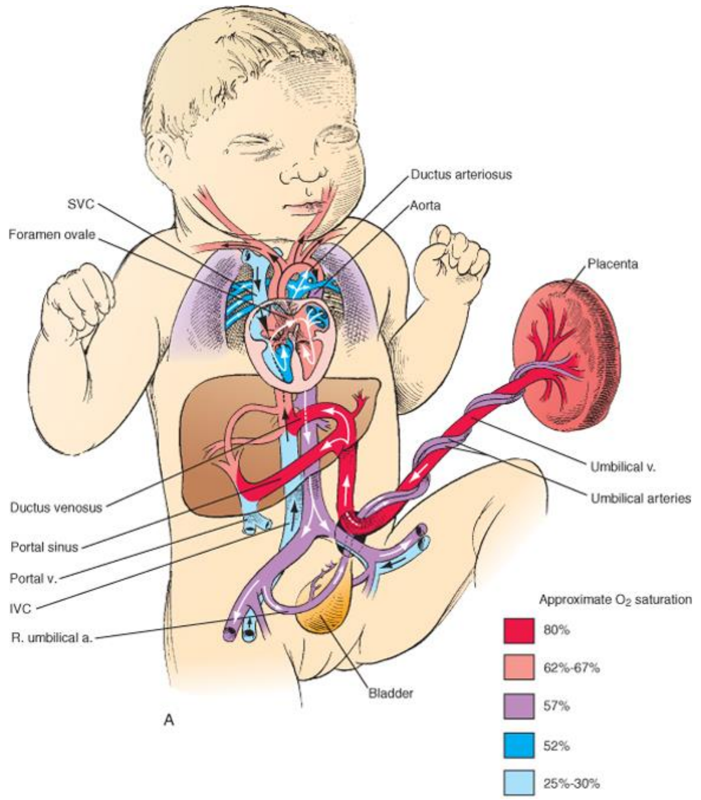


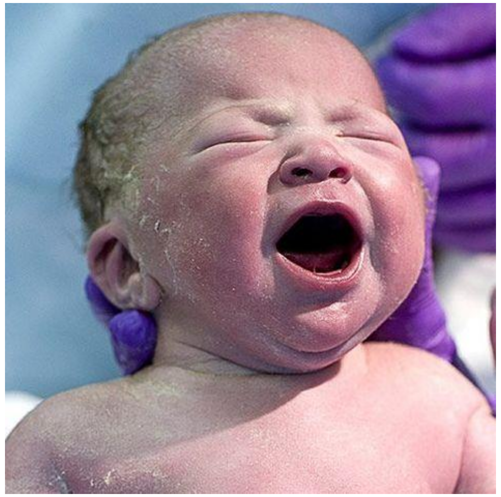




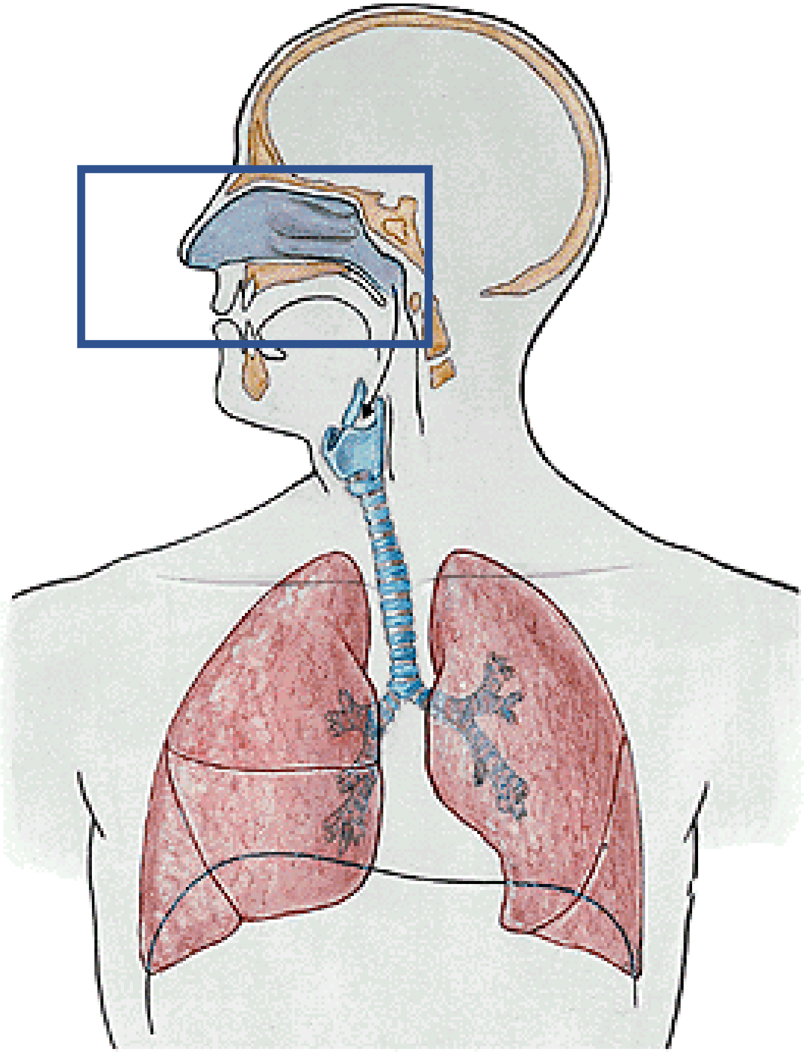
EM 15 Development of the respiratory system. Postpartum adaptation of the circulatory system

Dr. Krisztina Minkó

2026



Development of the nasal cavity



6th week: **nasal pits** deepen

oronasal membranes separates the nasal and oral cavities
primitive choanaes forms by its rupture on the 7th week

primary palate forms from posterior extension of intermaxillary processes

7th and 8th weeks: formation of **secondary palate** (from the maxillary prominences)
and **definitive choanae**

formation of **hard (intramembranous condensations) and soft palate** (from myogenic mesenchyme)

ectoderm and mesoderm of the frontonasal prominence and the medial nasal processes proliferate to form a midline nasal septum that grows down from the roof of the nasal cavity to fuse with the upper surface of the primary and secondary palates along the midline

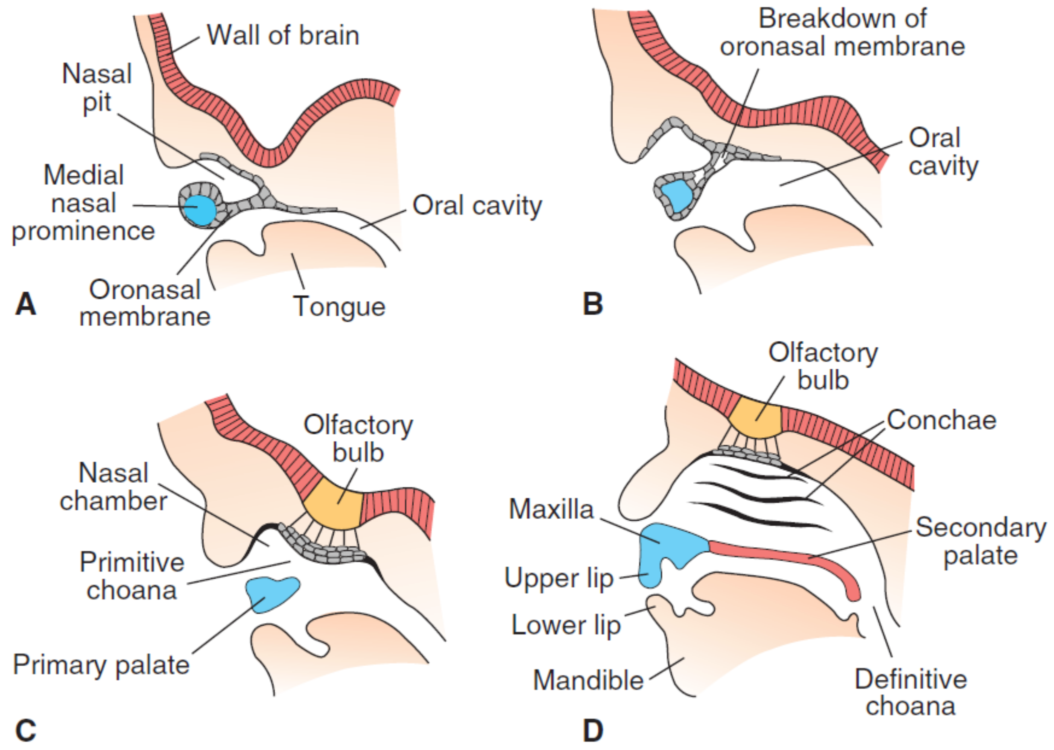
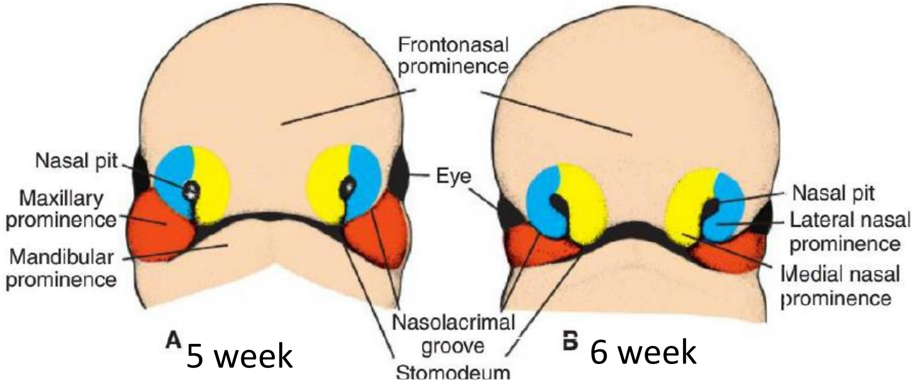


Figure 17.31 **A.** Sagittal section through the nasal pit and lower rim of the medial nasal prominence of a 6-week embryo. The primitive nasal cavity is separated from the oral cavity by the oronasal membrane. **B.** Similar section as in **A** showing the oronasal membrane breaking down. **C.** A 7-week embryo with a primitive nasal cavity in open connection with the oral cavity. **D.** Sagittal section through the face of a 9-week embryo showing separation of the definitive nasal and oral cavities by the primary and secondary palate. Definitive choanae are at the junction of the oral cavity and the pharynx.



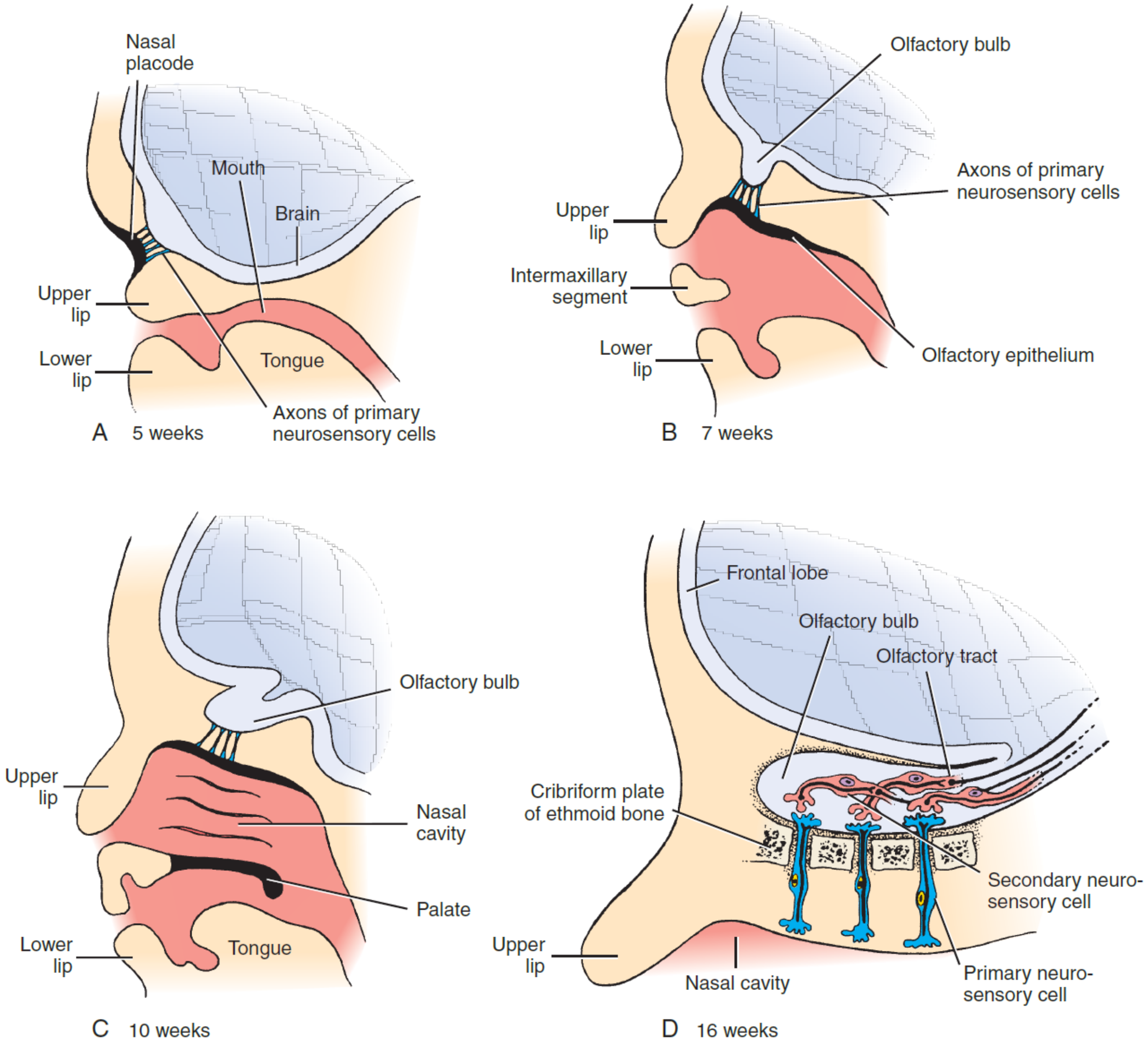
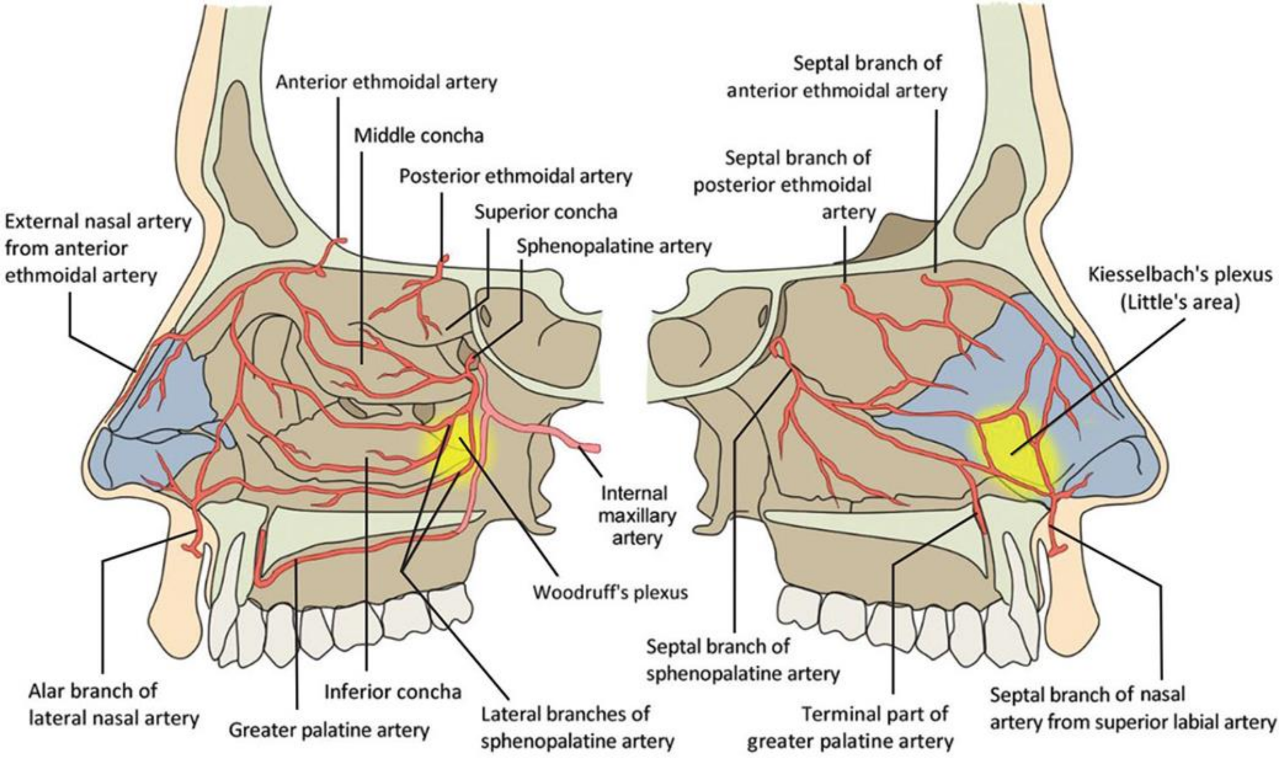
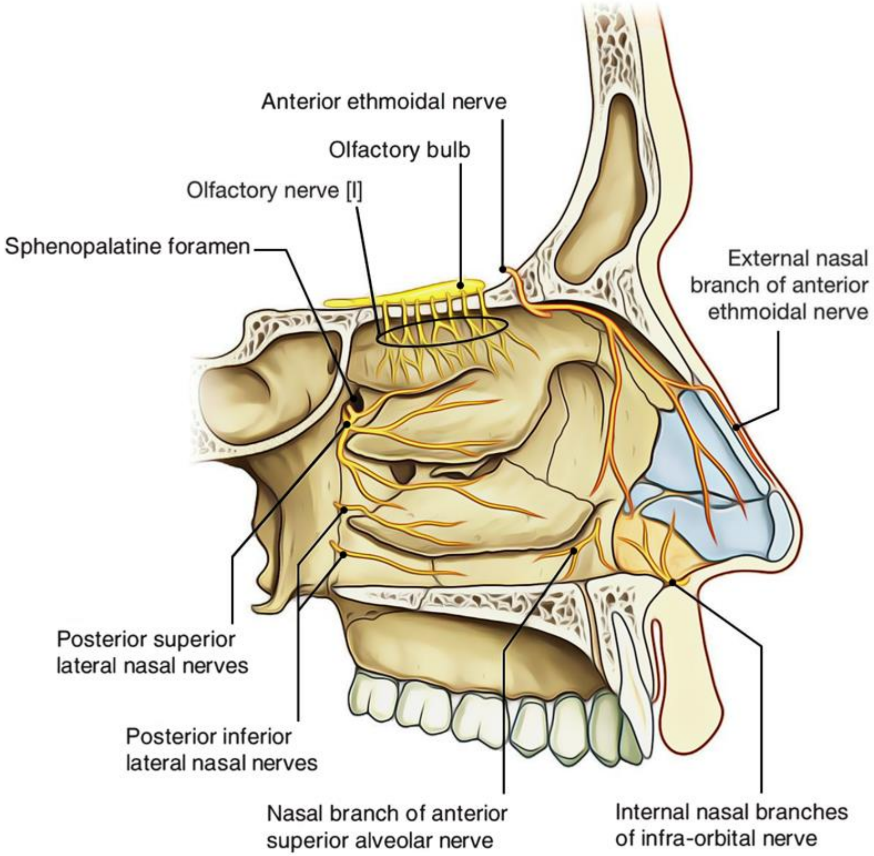


Figure 9-25. Formation of the olfactory tract as seen in sagittal views. *A*, During the 5th week, cells of the nasal placode differentiate into the primary neurosensory cells of the olfactory tract and produce axons that grow into the presumptive olfactory bulb of the adjacent telencephalon. There they synapse there with secondary neurons. *B-D*, As development continues, the elongating axons of the secondary olfactory neurons in the olfactory bulb produce the olfactory tract.

Dual origin of blood supply and innervation of the nasal cavity refers to its different embryonic origins





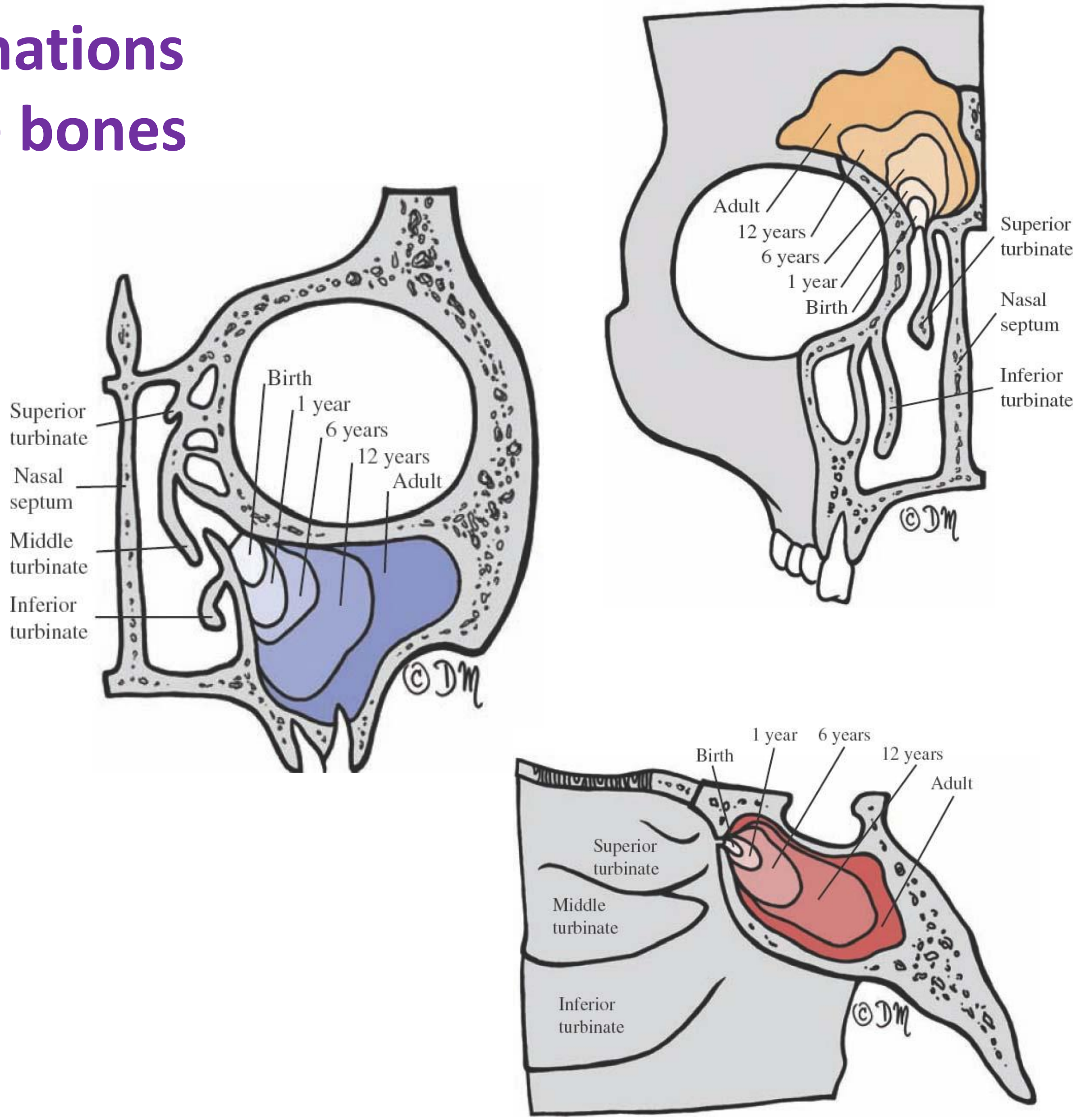
Paranasal sinuses develop from invaginations of the nasal cavity that extend into the bones

Two of the sinuses appear during fetal life (maxillary and ethmoid) and the other two (sphenoid and frontal) appear after birth.

The resulting cavities are small at birth but expand throughout childhood.

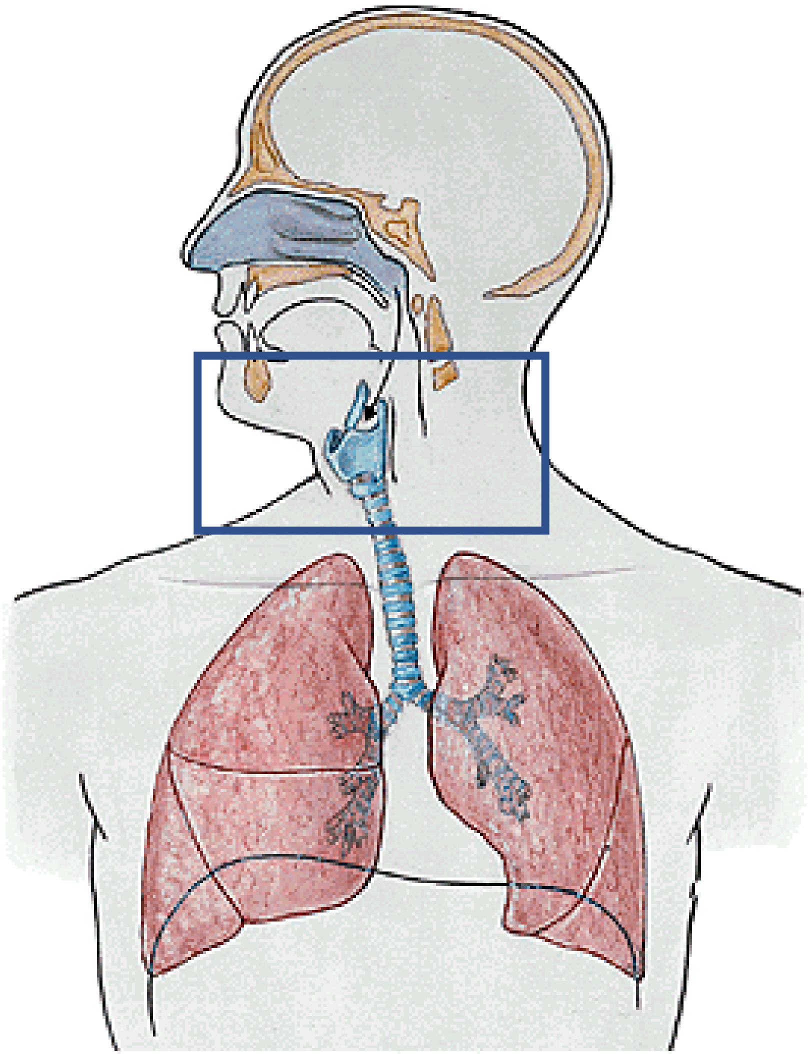
These sinuses do not complete their growth until puberty.

The frontal sinuses do not form until the 5th or 6th postnatal year.



https://entokey.com/surgical-anatomy-of-the-paranasal-sinus/

Development of the larynx



Epithelium mucosae of the larynx develops from the endoderm

Laryngotracheal orifice changes from sagittal slit to a T-shaped opening

hypobranchial eminence is the anterior border

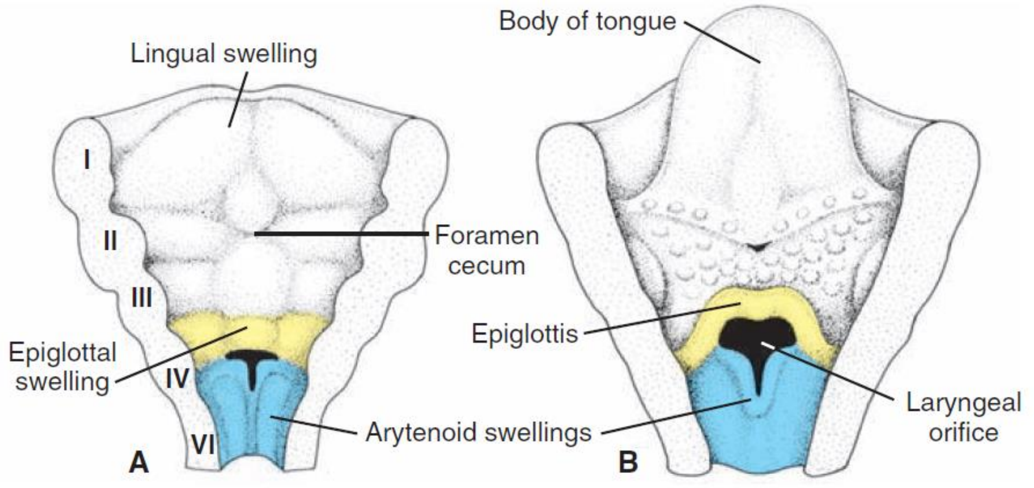
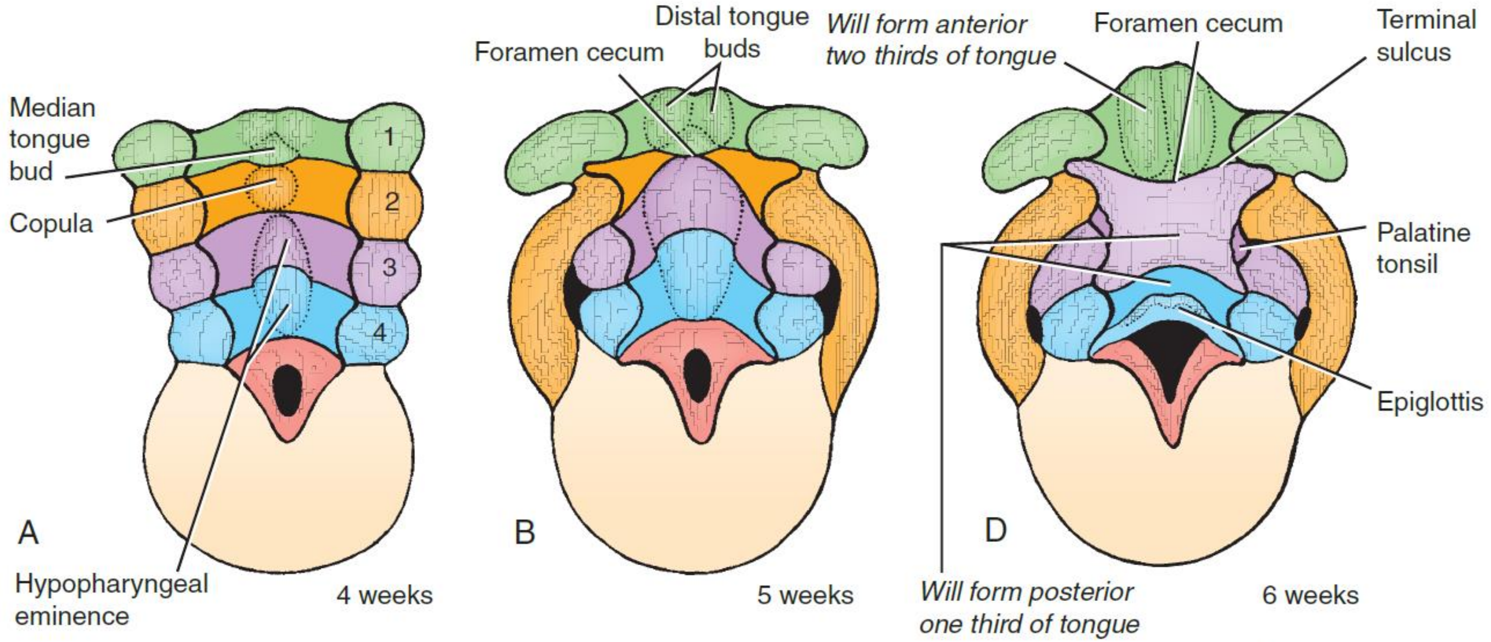


Figure 14.4 Laryngeal orifice and surrounding swellings at successive stages of development. **A.** 6 weeks. **B.** 12 weeks.



1. Epithelium proliferates: **temporary occlusion of the lumen**

2. After **recanalisation** the laryngeal ventricles form

3. Formation of **false and true vocal cords**

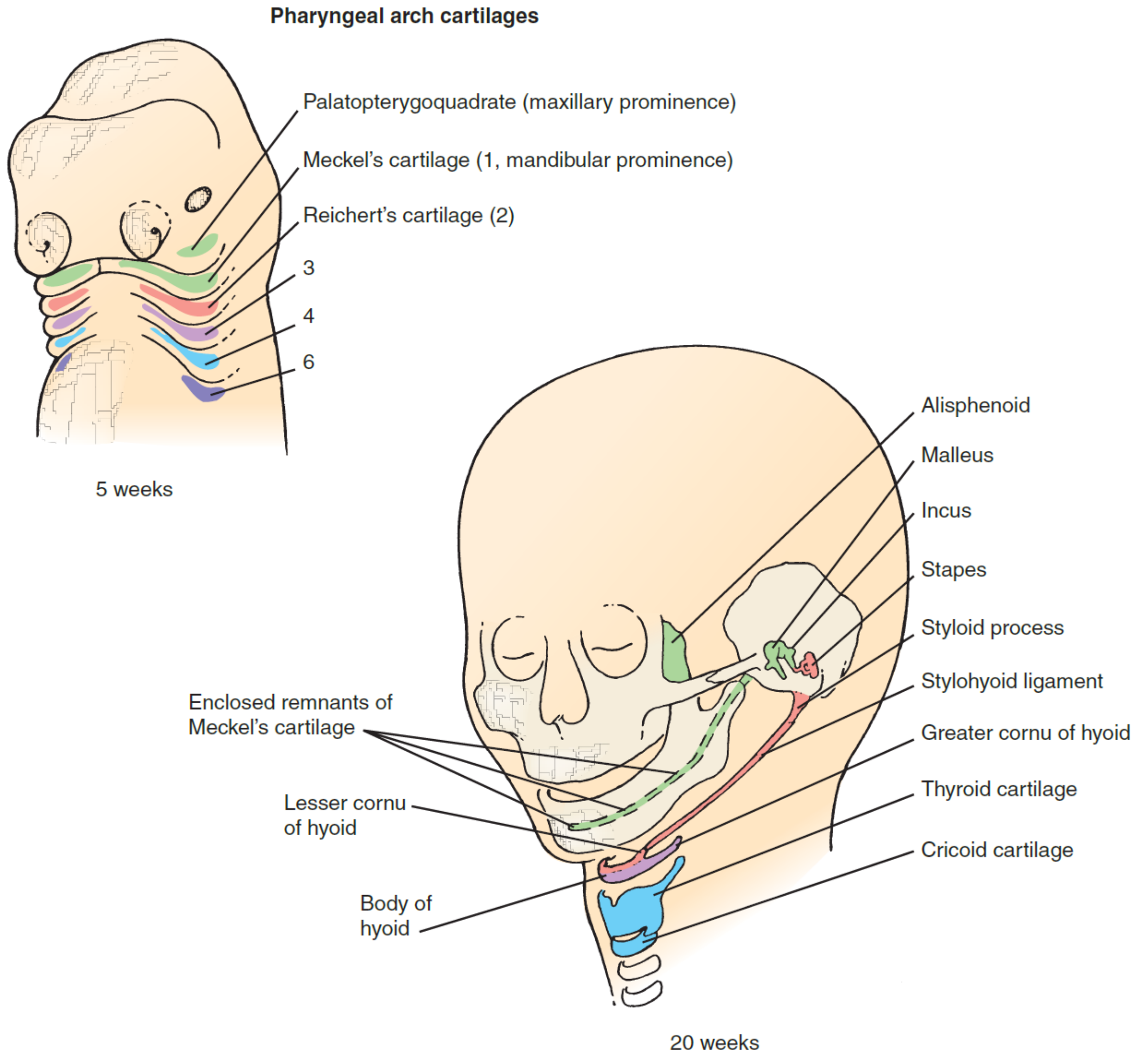
Cartilages, muscles of the larynx originate from the mesencyme of the 4th and 6th pharyngeal arches

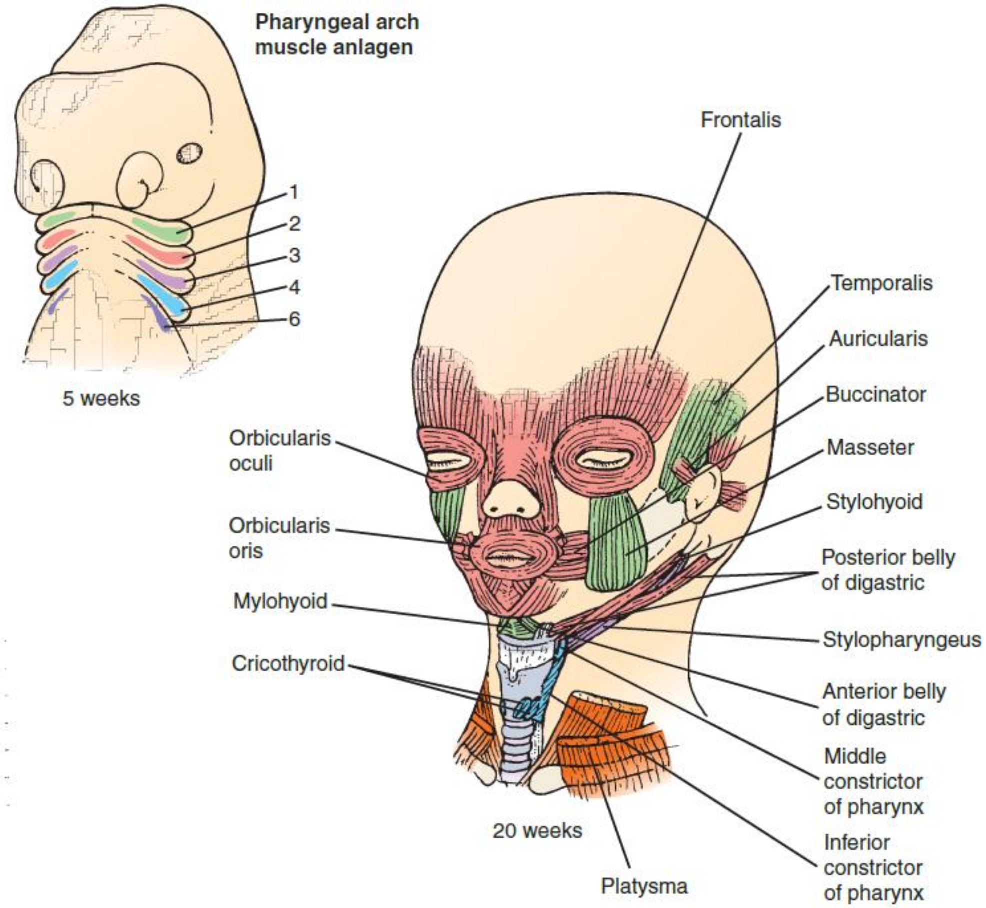
Table 16-1 Derivatives of the Pharyngeal Arches and Their Tissues of Origin

Pharyngeal Arch	Arch Artery[a]	Skeletal Elements	Muscles	Cranial Nerve[b]
1	Terminal branch of maxillary artery	Derived from arch cartilages (originating from neural crest cells): From maxillary cartilage: alisphenoid, incus From Meckel's cartilage: malleus Derived by direct ossification from arch dermal mesenchyme: maxilla, zygomatic, squamous portion of temporal bone, mandible (originate from neural crest cells)	Muscles of mastication (temporalis, masseter, and pterygoids), mylohyoid, anterior belly of the digastric, tensor tympani, tensor veli palatini (originate from head mesoderm)	Maxillary and mandibular division of trigeminal nerve (V)
2	Stapedial artery (embryonic), corticotympanic artery (adult)	Stapes, styloid process, lesser horns and upper rim of hyoid (derived from the second-arch [Reichert's] cartilage; originate from neural crest cells)	Muscles of facial expression (orbicularis oculi, orbicularis oris, risorius, platysma, auricularis, frontalis, and buccinator), posterior belly of the digastric, stylohyoid, stapedius (originate from head mesoderm)	Facial nerve (VII)
3	Common carotid artery, root of internal carotid	Lower rim and greater horns of hyoid (derived from the third-arch cartilage; originate from neural crest)	Stylopharyngeus (originates from head mesoderm)	Glossopharyngeal nerve (IX)
4	Arch of aorta (left side), right subclavian artery (right side); original sprouts of pulmonary arteries	Laryngeal cartilages (derived from the fourth-arch cartilage; originate from neural crest cells)	Constrictors of pharynx, cricothyroid, levator veli palatini (originate from occipital somites 2 to 4)	Superior laryngeal branch of vagus nerve (X)
6	Ductus arteriosus; roots of definitive pulmonary arteries	Laryngeal cartilages (derived from the sixth-arch cartilage; originate from neural crest cells)	Intrinsic muscles of larynx (originate from occipital somites 1 and 2)	Recurrent laryngeal branch of vagus nerve (X)

The fourth and sixth pharyngeal arches together give rise to the larynx, consisting of the thyroid, cuneiform, corniculate, arytenoid, and cricoid cartilages.

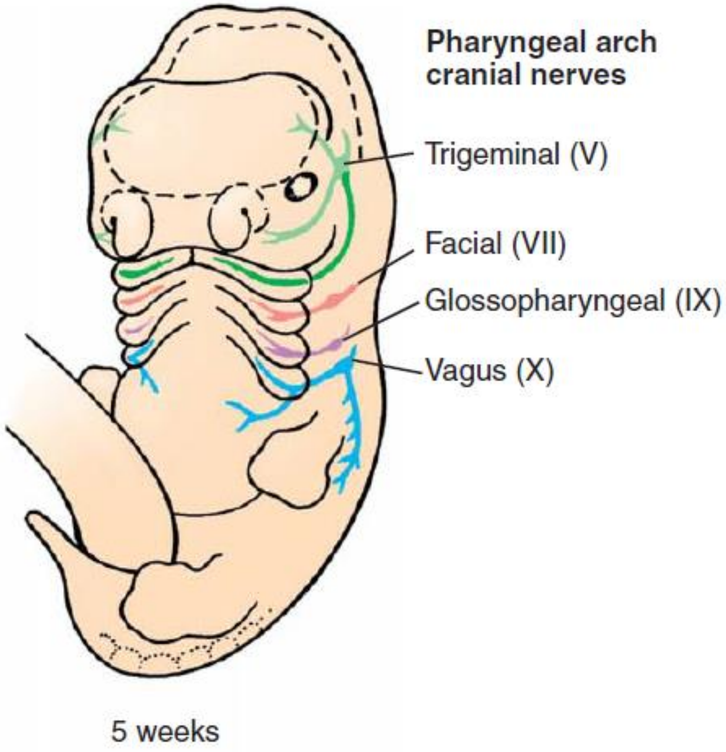
Recent neural crest cell fate mapping studies in mice have shown that these **cartilages arise from neural crest cells**, with the mesodermal boundary forming at the tracheal cartilages.
The epiglottal cartilages do not form until the 5th month, long after the other pharyngeal arch cartilages have formed. These cartilages arise in the fourth-arch area, but their precise origin is unknown.



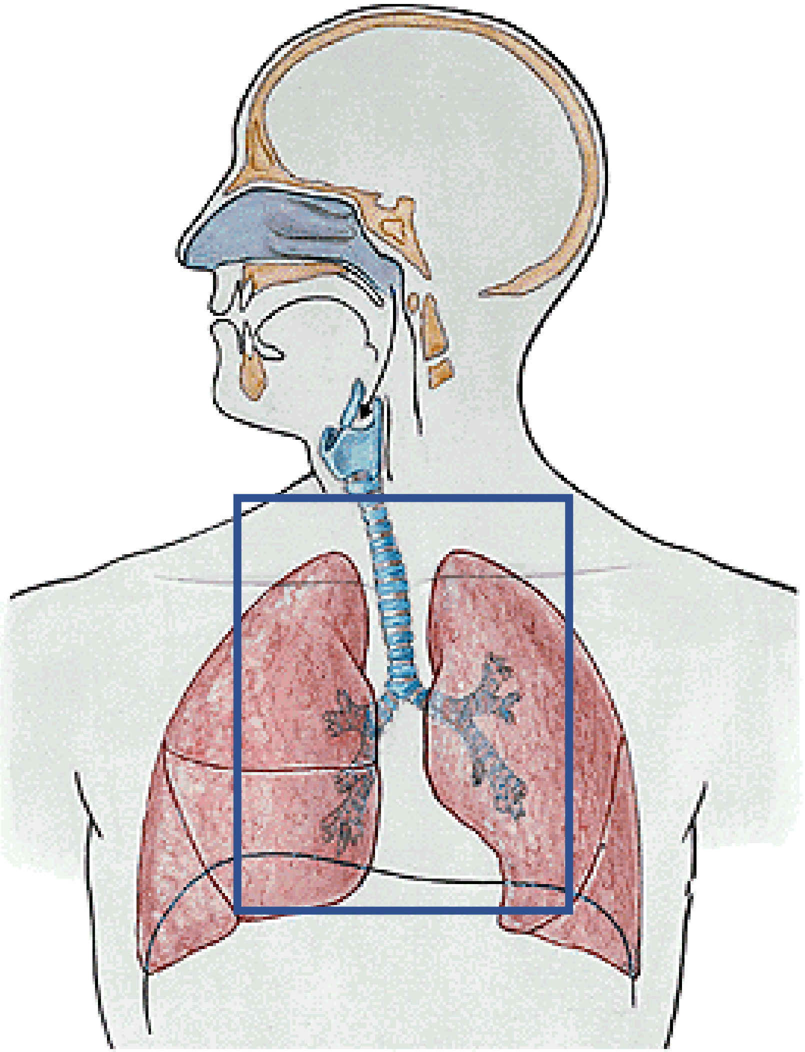


Muscles of the fourth arch (**cricothyroid, levator palatini**, and **constrictors of the pharynx**) are innervated by the **superior laryngeal branch of the vagus**, the nerve of the fourth arch.
Intrinsic muscles of the larynx are supplied by the **recurrent laryngeal branch of the vagus**, the nerve of the sixth arch.

The muscles that form in each pharyngeal arch are innervated by a cranial nerve branch that is specific to that arch, and they maintain their relationship in the adult. This close relationship has been conserved since the evolution of jawed fishes; along with the pharyngeal pouches, it defines a conserved segmental organization for the pharyngeal arch system.



Development of the lower airways, trachea, bronchi, lungs



The lung is a composite of endodermal and mesodermal tissues.

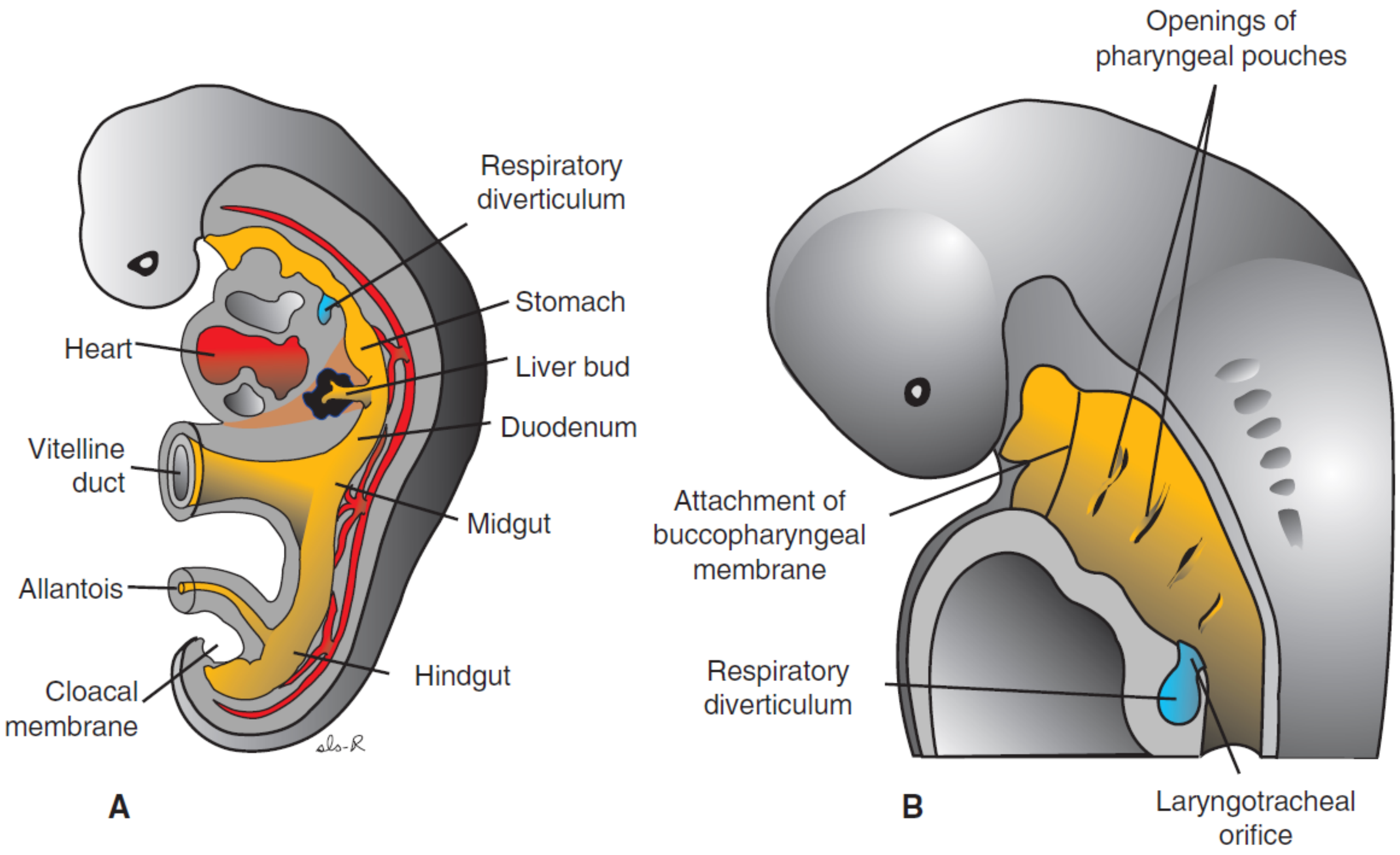
The **endoderm of the respiratory diverticulum** gives rise to the mucosal lining of the bronchi and to the epithelial cells of the alveoli.

The remaining components of the lung, including the muscle and cartilage supporting the bronchi and the visceral pleura covering the lung, are derived from the **splanchnopleuric mesoderm**, which covers the bronchi as they grow out from the mediastinum into the pleural space.

The lung vasculature is thought to develop via **angiogenesis** (i.e., sprouting from neighboring vessels).

Respiratory diverticulum is a ventral evagination of the foregut

On day 22 the foregut produces a ventral evagination called the **respiratory diverticulum or lung bud**, which is the primordium of the lungs.



4th week

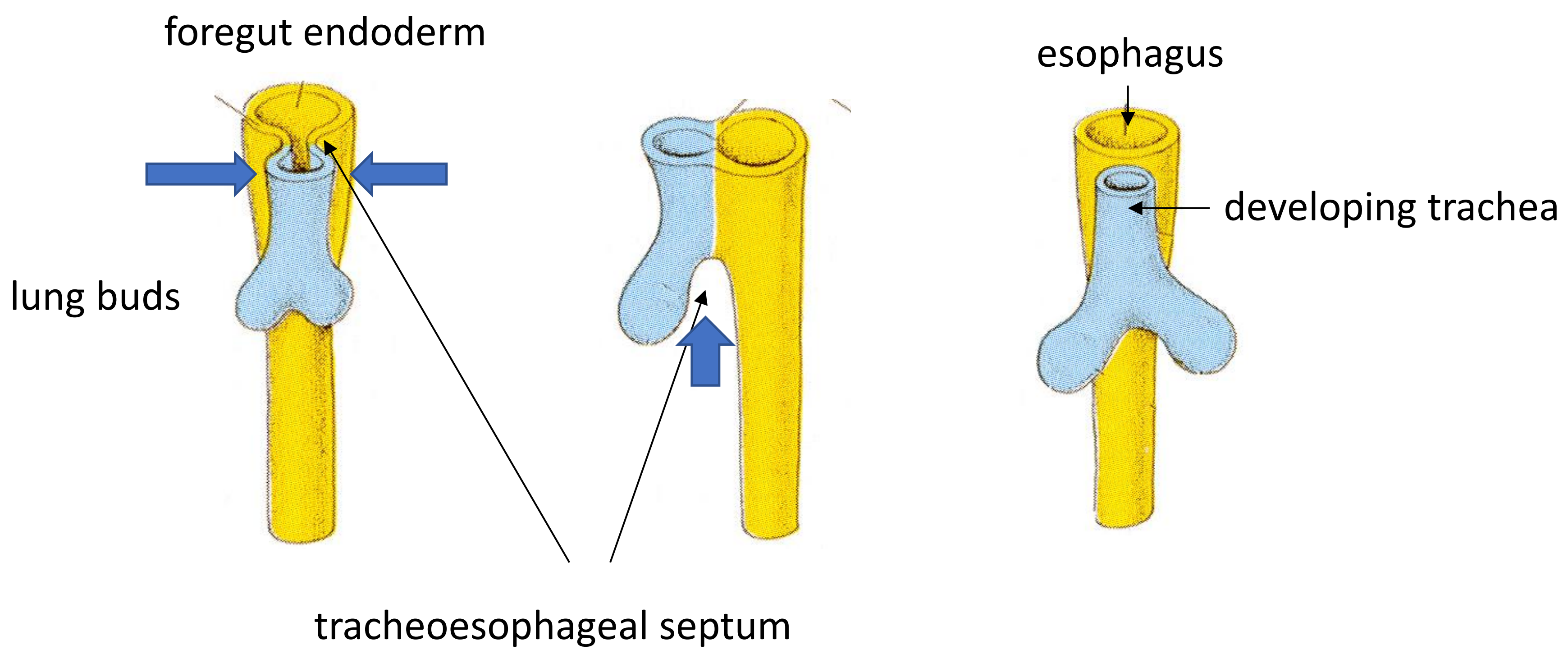
Figure 14.1 **A.** Embryo of approximately 25 days' gestation showing the relation of the respiratory diverticulum to the heart, stomach, and liver. **B.** Sagittal section through the cephalic end of a 5-week embryo showing the openings of the pharyngeal pouches and the laryngotracheal orifice.

Respiratory diverticulum grows ventrocaudally through the mesenchyme surrounding the foregut

on days 26 to 28, it undergoes a first bifurcation, splitting into right and left primary bronchial (or lung) buds.

These buds are the rudiments of the two lungs and the right and left primary bronchi, and the proximal end (stem) of the diverticulum forms the trachea and larynx. The latter opens into the pharynx via the glottis.

tracheoesophageal septum grows between esophagus and trachea



As the primary bronchial buds form, the stem of the diverticulum begins to separate from the overlying portion of the pharynx, which becomes the esophagus.

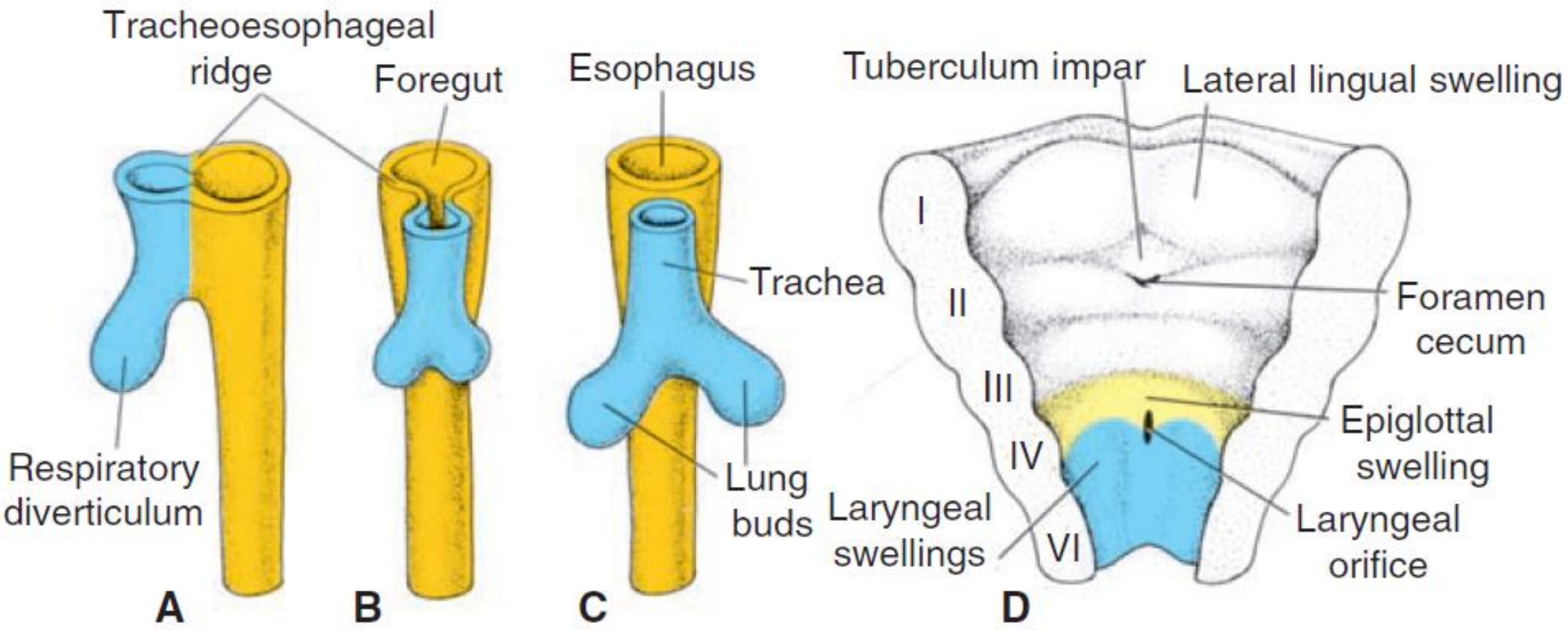


Figure 14.2 A–C. Successive stages in development of the respiratory diverticulum showing the tracheoesophageal ridges and formation of the septum, splitting the foregut into esophagus and trachea with lung buds. **D.** The ventral portion of the pharynx seen from above showing the laryngeal orifice and surrounding swelling.

tracheoesophageal septum from lateral view

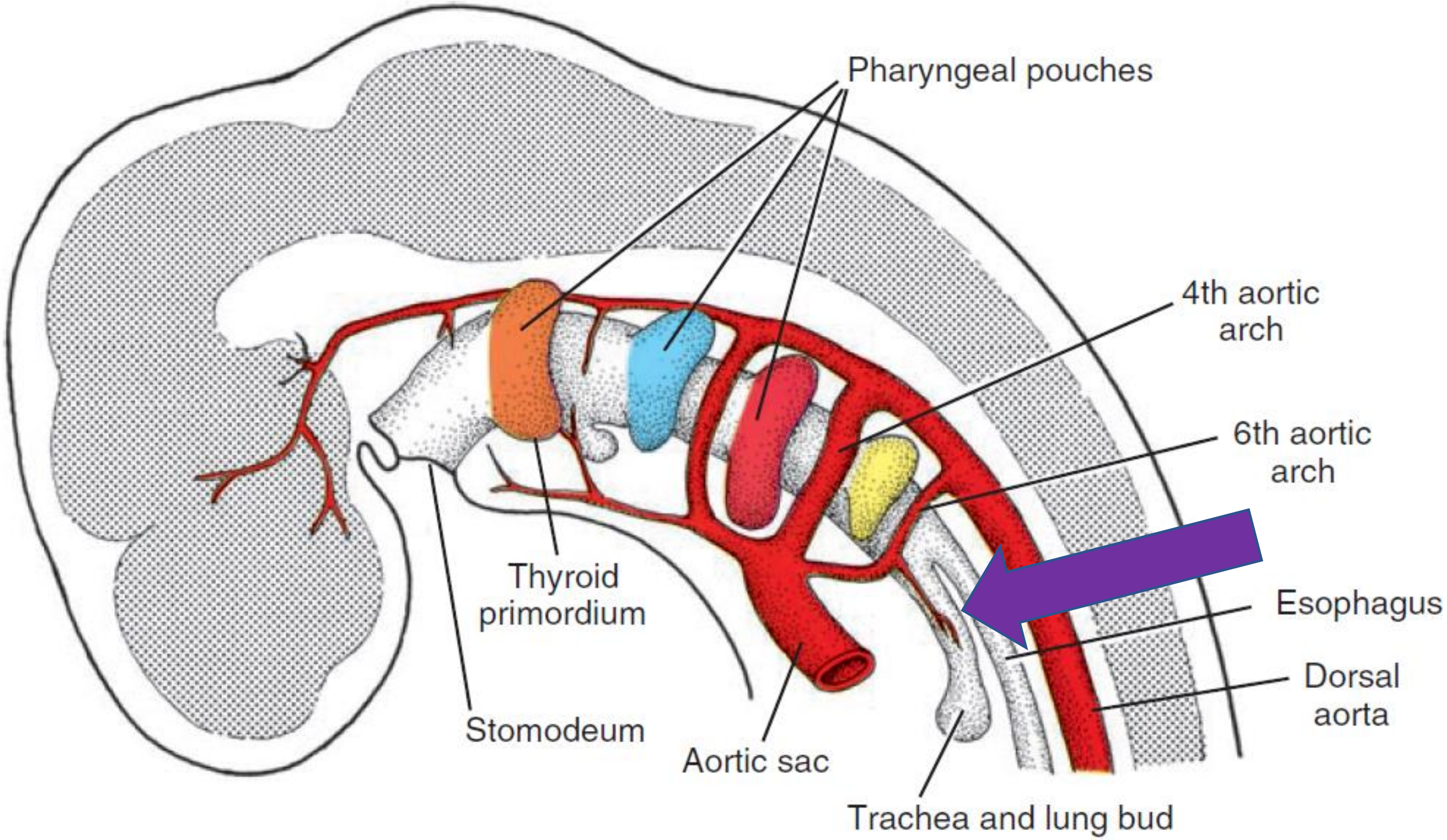


Figure 17.4 Pharyngeal pouches as outpocketings of the foregut and the primordium of the thyroid gland and aortic arches.

Abnormalities in partitioning of the esophagus and trachea by the tracheoesophageal septum result in esophageal atresia with or without tracheoesophageal fistula

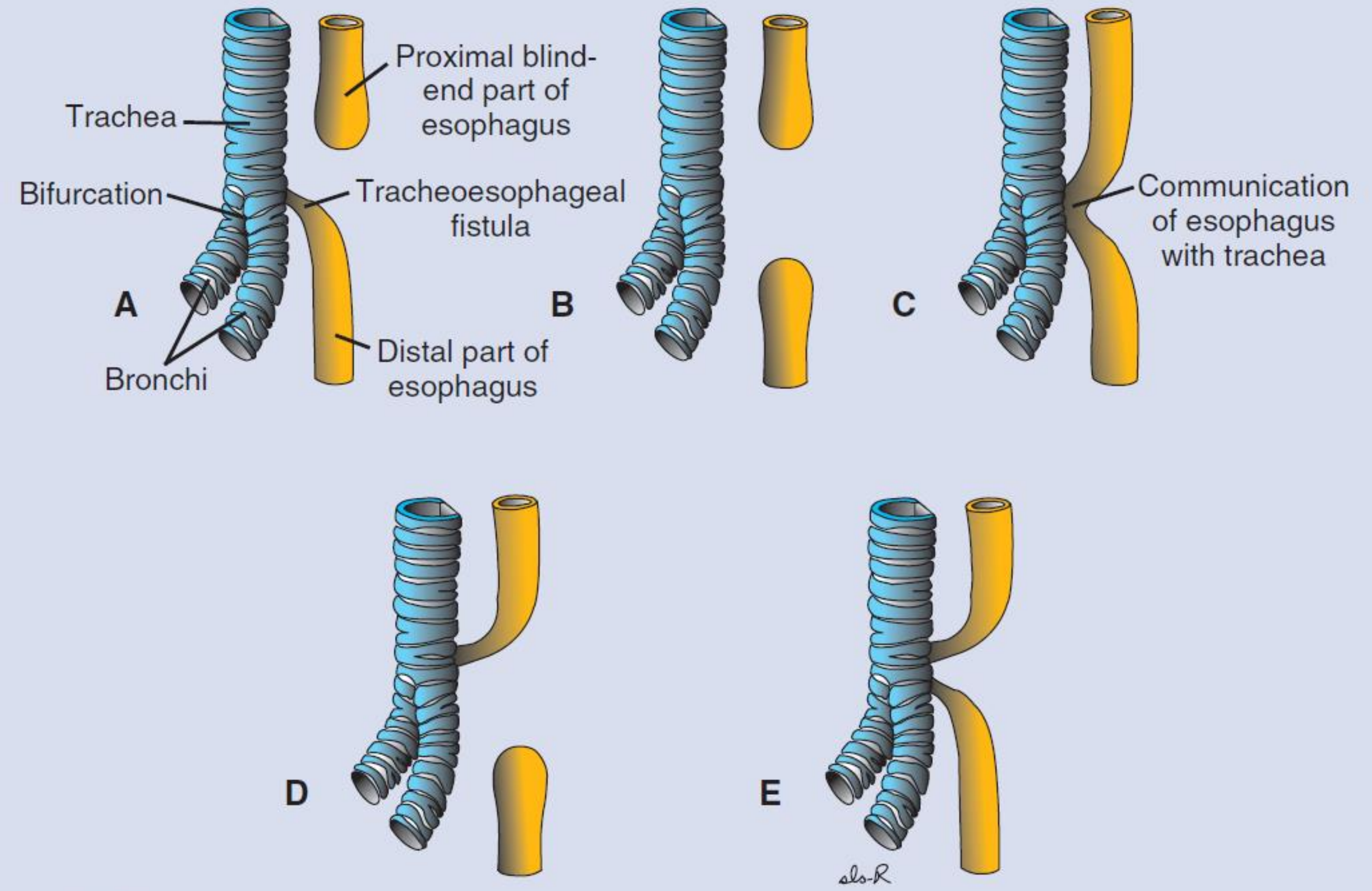


Figure 14.3 Various types of esophageal atresia and/or TEFs. **A.** The most frequent abnormality (90% of cases) occurs with the upper esophagus ending in a blind pouch and the lower segment forming a fistula with the trachea. **B.** Isolated esophageal atresia (4% of cases). **C.** H-type TEF (4% of cases). **D,E.** Other variations (each 1% of cases).

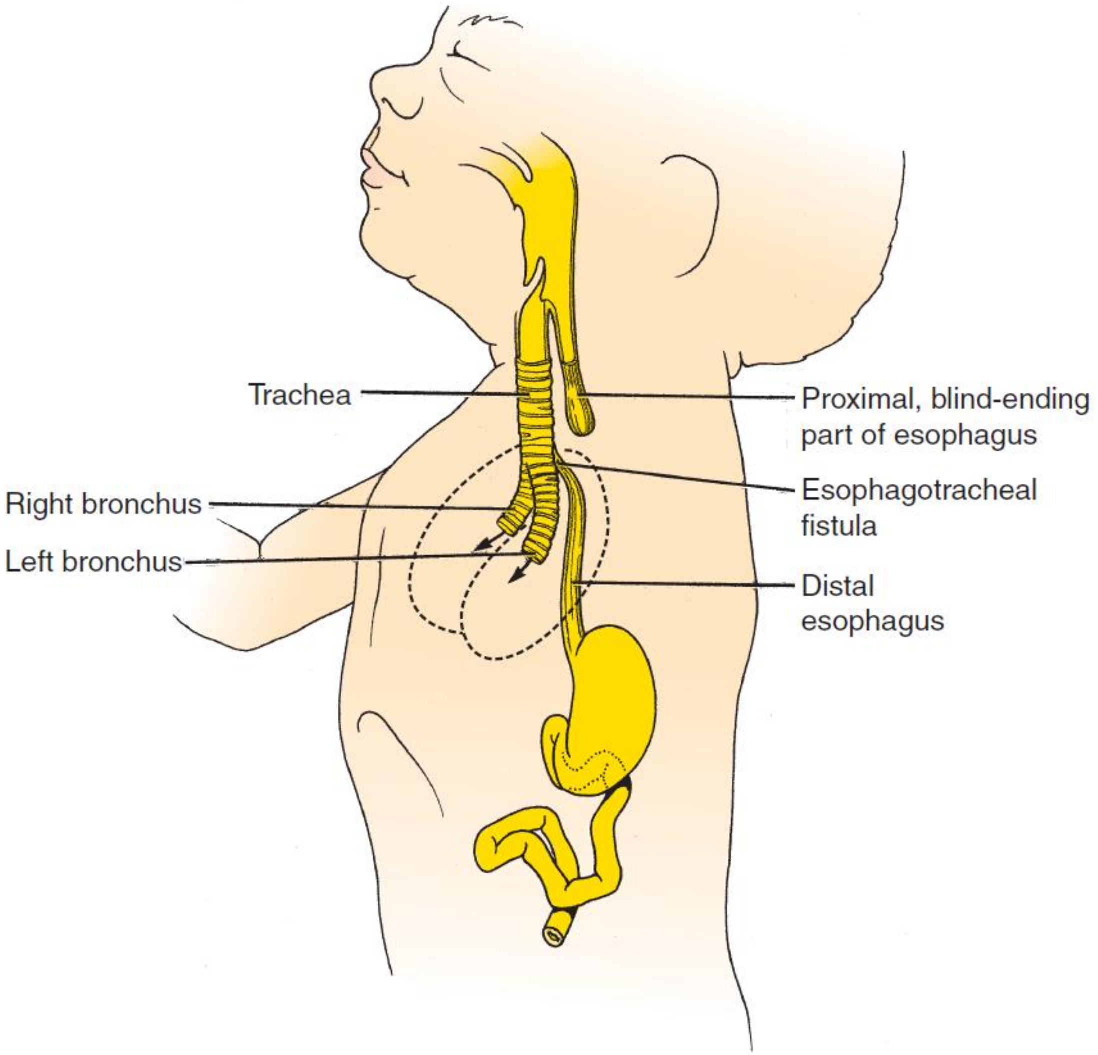
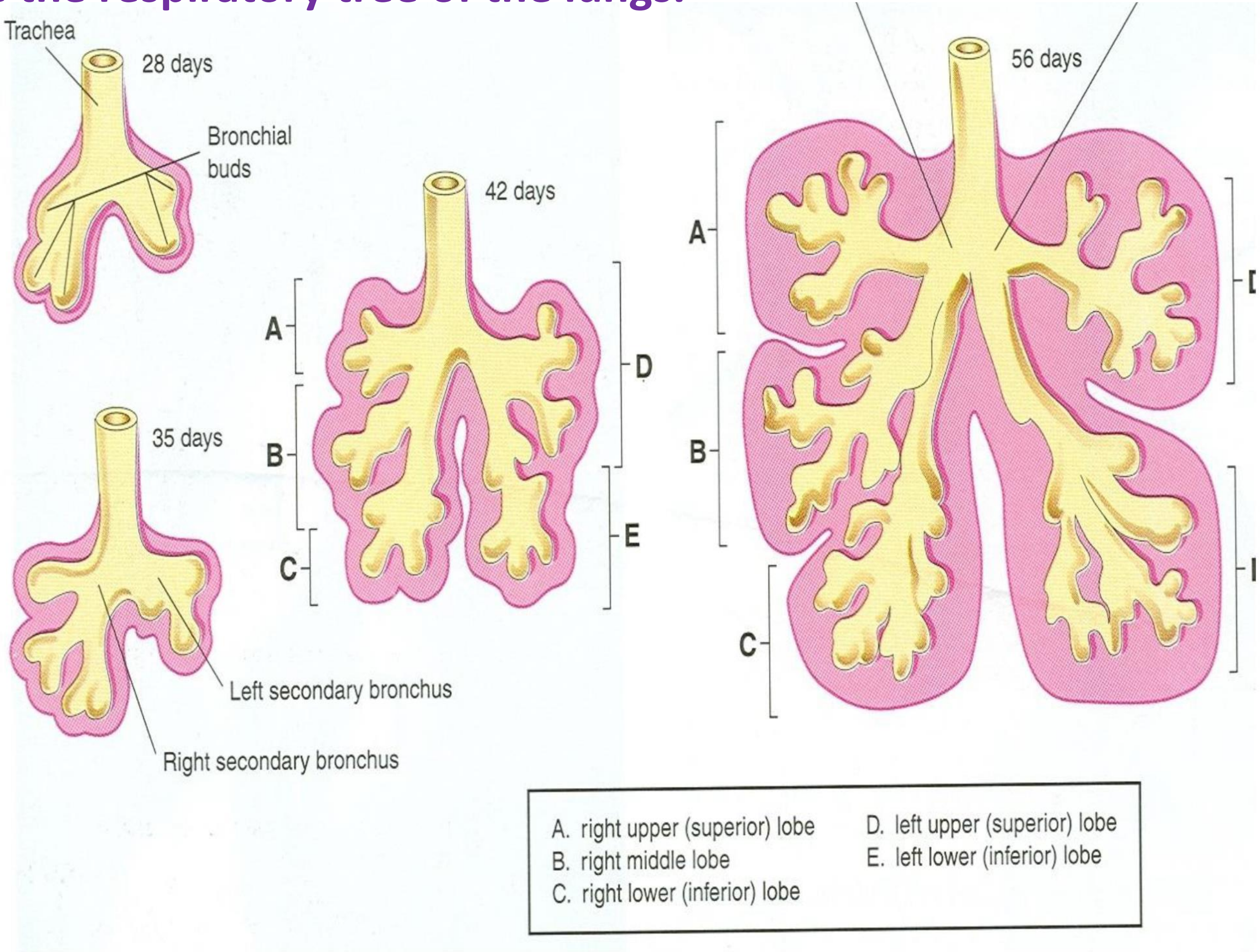


Figure 11-4. Diagram of an infant with esophageal atresia and tracheoesophageal fistula shows how the first drink of fluid after birth could be diverted into the newly expanded lungs (arrows).

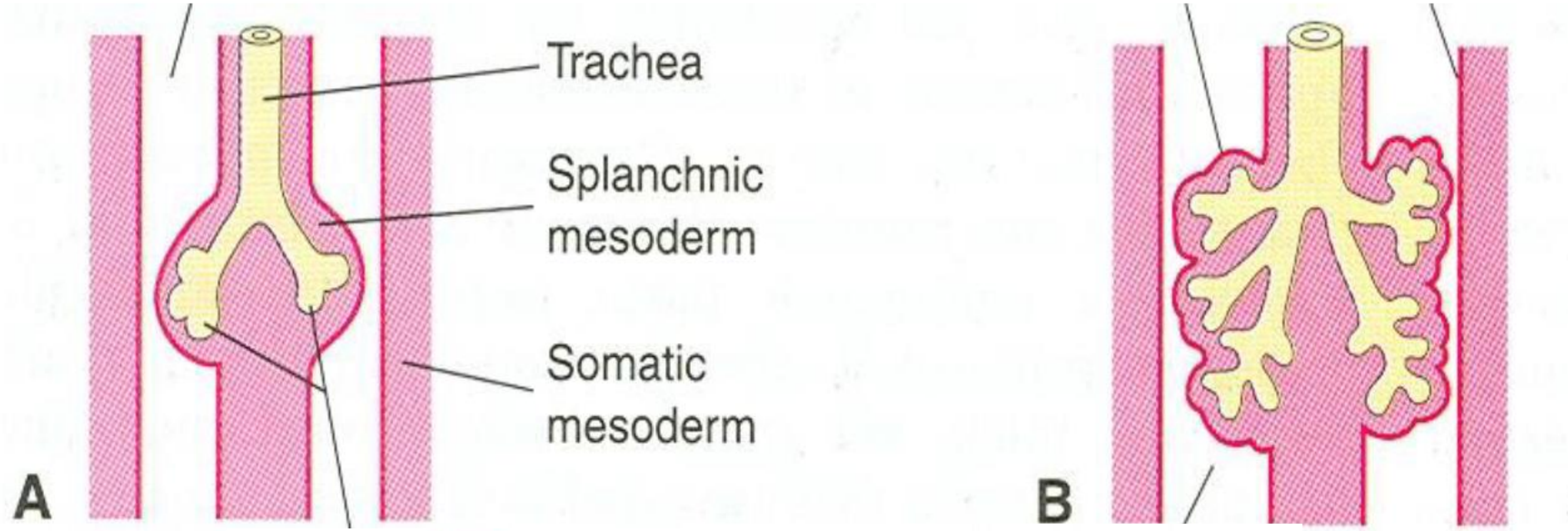
During weeks 5 and 28, the primary bronchial buds undergo about 16 rounds of branching to generate the respiratory tree of the lungs.

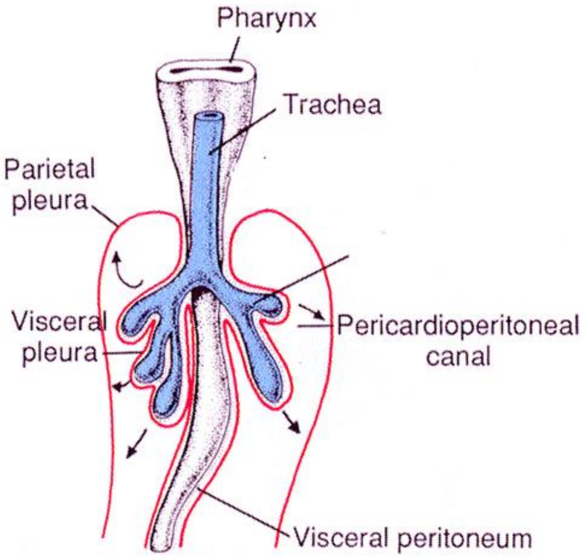
The pattern of branching of the lung endoderm is regulated by the surrounding mesenchyme, which invests the buds from the time that they first form.

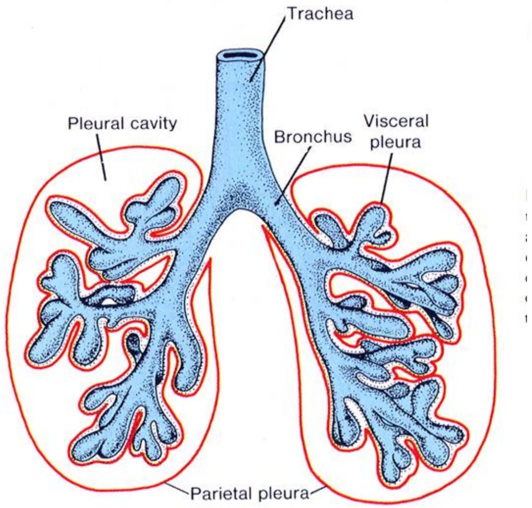
The first round of branching of the primary bronchial buds occurs early in the 5th week. This round of branching is highly stereotypical and yields three secondary bronchial buds on the right side and two on the left.
The secondary bronchial buds give rise to the lung lobes: three in the right lung and two in the left lung.
During the 6th week, a more variable round of branching typically yields 10 tertiary bronchi on both sides; these become the **bronchopulmonary segments** of the mature lung.



Pleura development: from splanchnic and somatic mesoderm

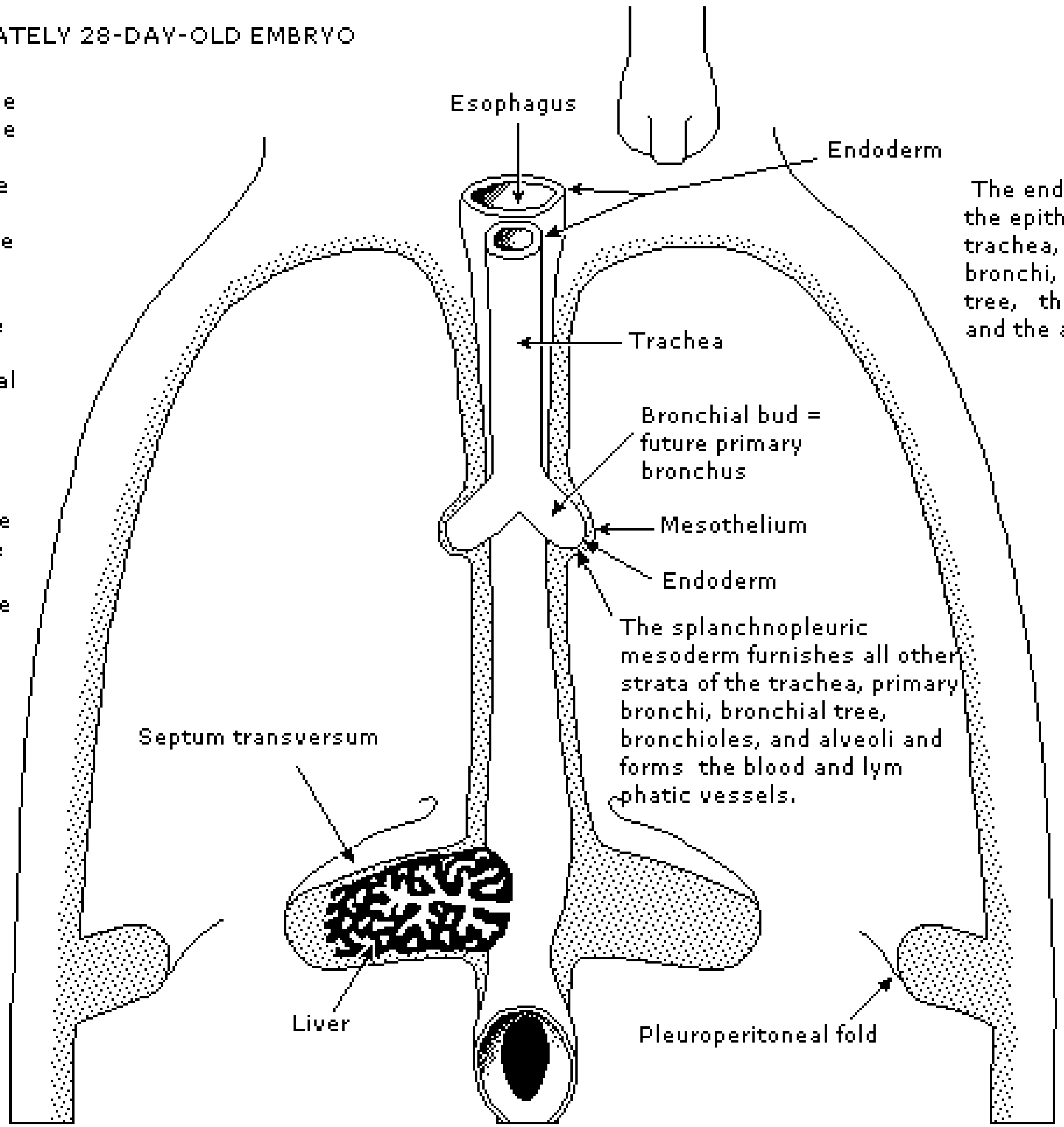






AN APPROXIMATELY 28-DAY-OLD EMBRYO

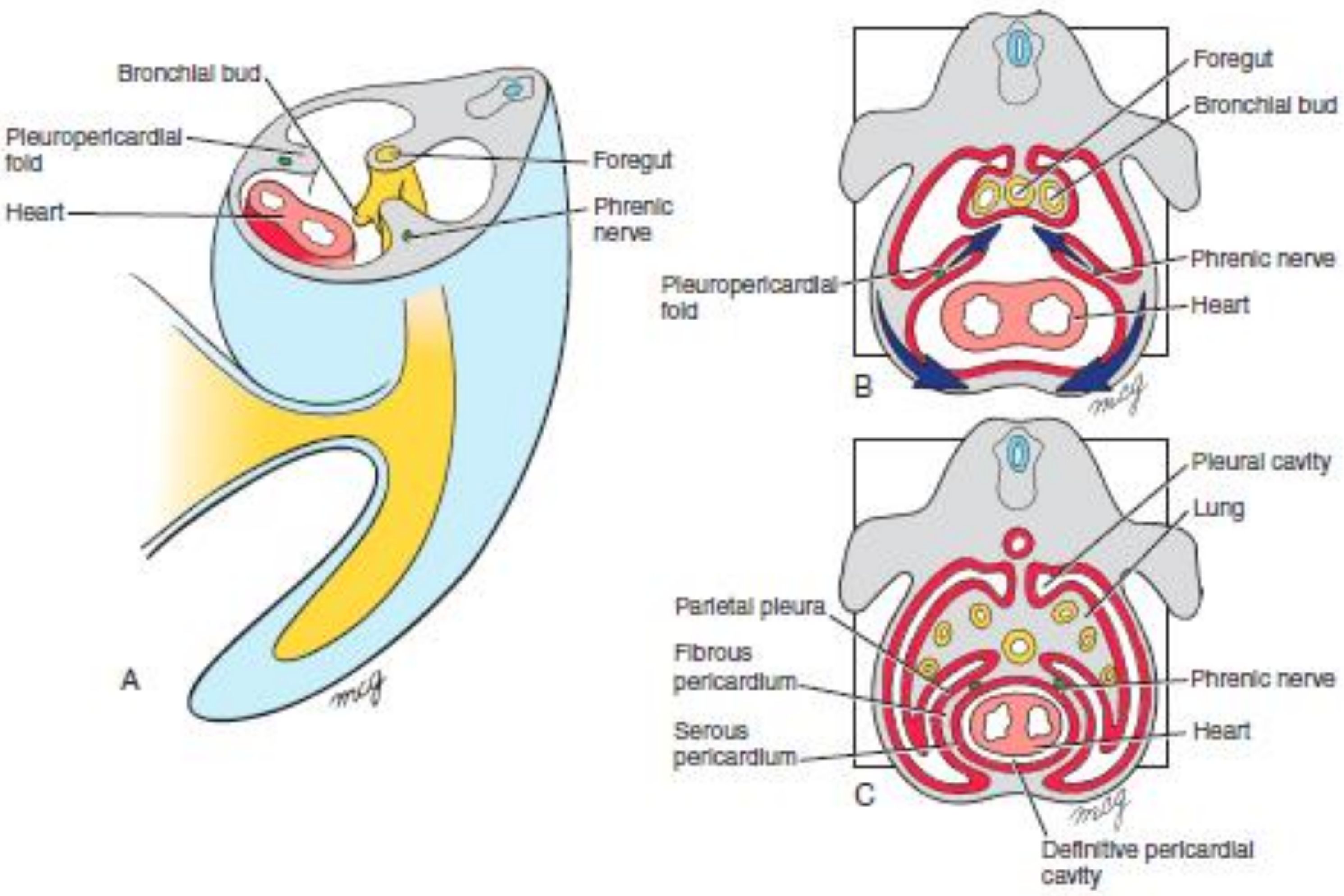
The trachea has separated from the esophagus and the former starts to branch to form the bronchial buds which will give rise to the primary bronchi. They penetrate into the corresponding pericardiperitoneal canals. Each bronchial bud is lined by the splanchnopleuric mesoderm and the mesothelium. The pleuroperitoneal folds approach the septum transversum in which the liver develops.



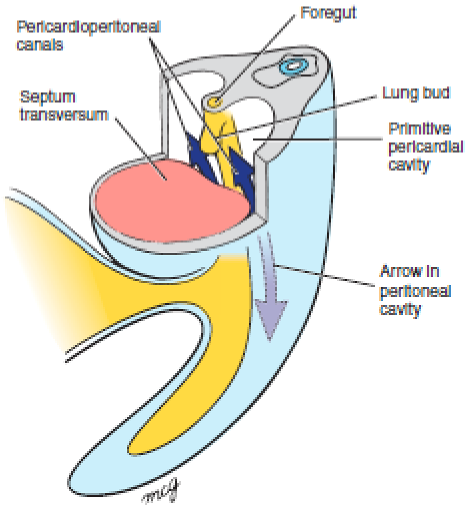
The endoderm furnishes the epithelium of the trachea, the primary bronchi, the bronchial tree, the bronchioles and the alveoli.

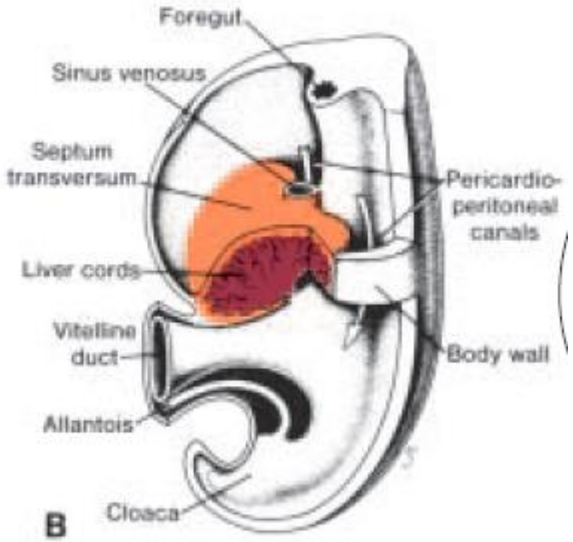
The splanchnopleuric mesoderm furnishes all other strata of the trachea, primary bronchi, bronchial tree, bronchioles, and alveoli and forms the blood and lym phatic vessels.

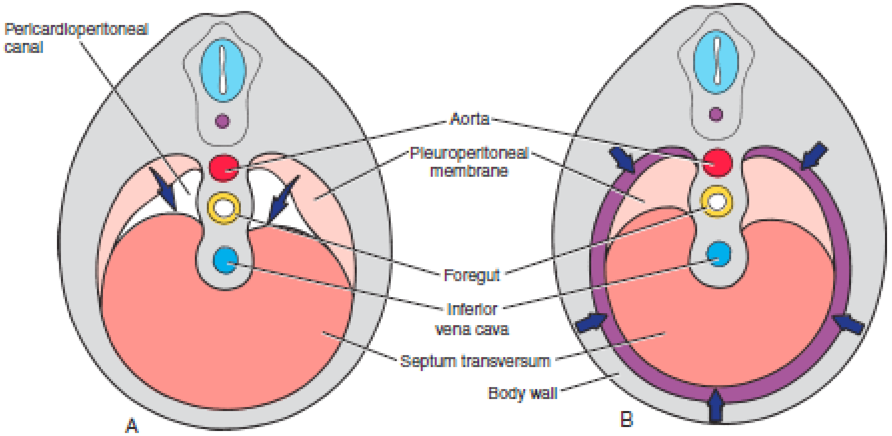
Septum pleuropericardialis



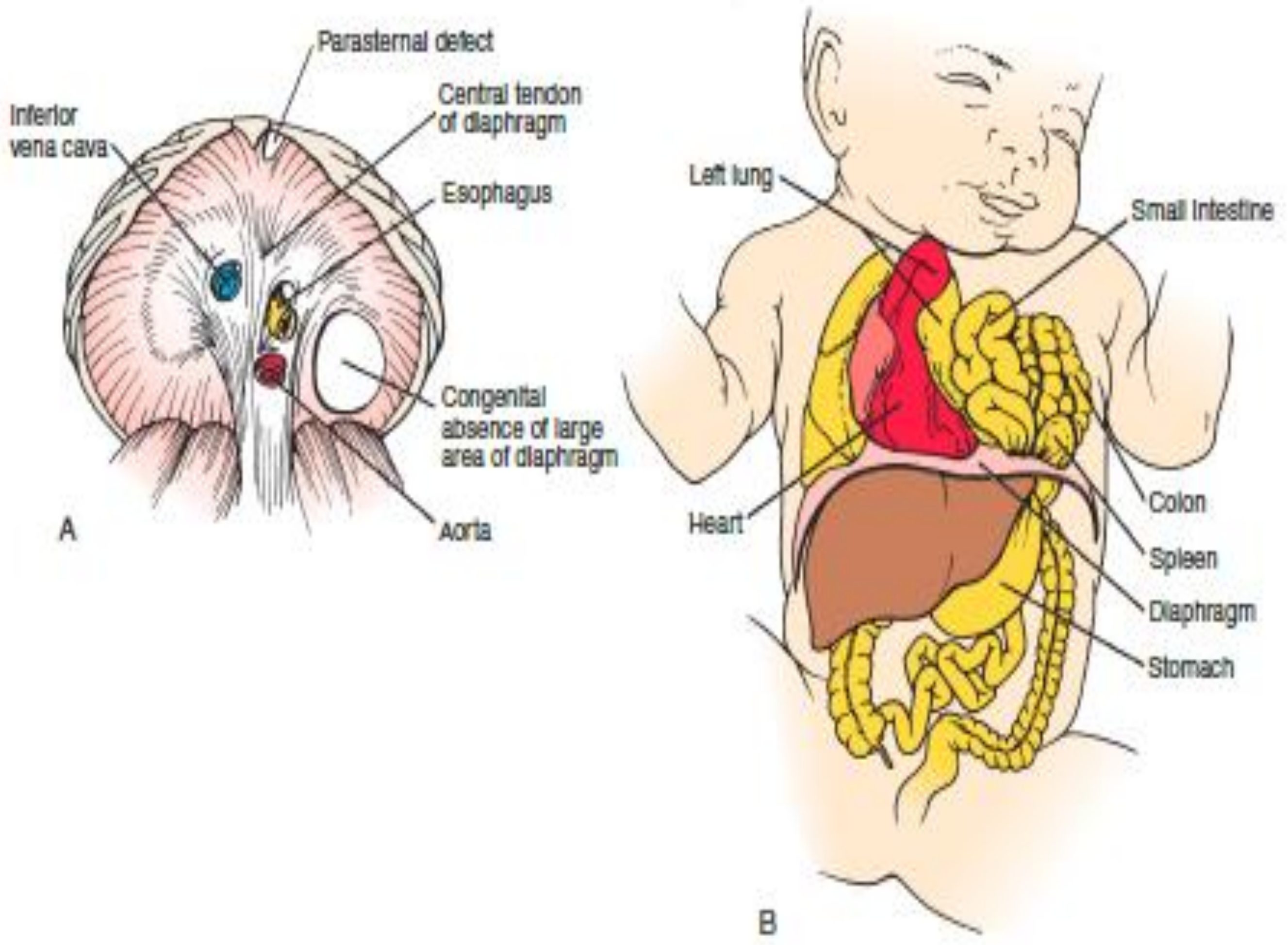
Septum pleuroperitonealis







Congenital hernia of the diaphragm



By week 16, after about 14 more branchings, the respiratory tree produces small branches called **terminal bronchioles.**

Between 16 and 28 weeks each terminal bronchiole divides into two or more **respiratory bronchioles**, and the mesodermal tissue surrounding these structures **becomes highly vascularized.**

By week 28 the respiratory bronchioles begin to sprout a final generation of stubby branches. These branches develop in craniocaudal progression,
forming first at more cranial terminal bronchioles.

By week 36 the first-formed wave of terminal branches are invested in a dense network of capillaries
and are called **terminal sacs (primitive alveoli).**

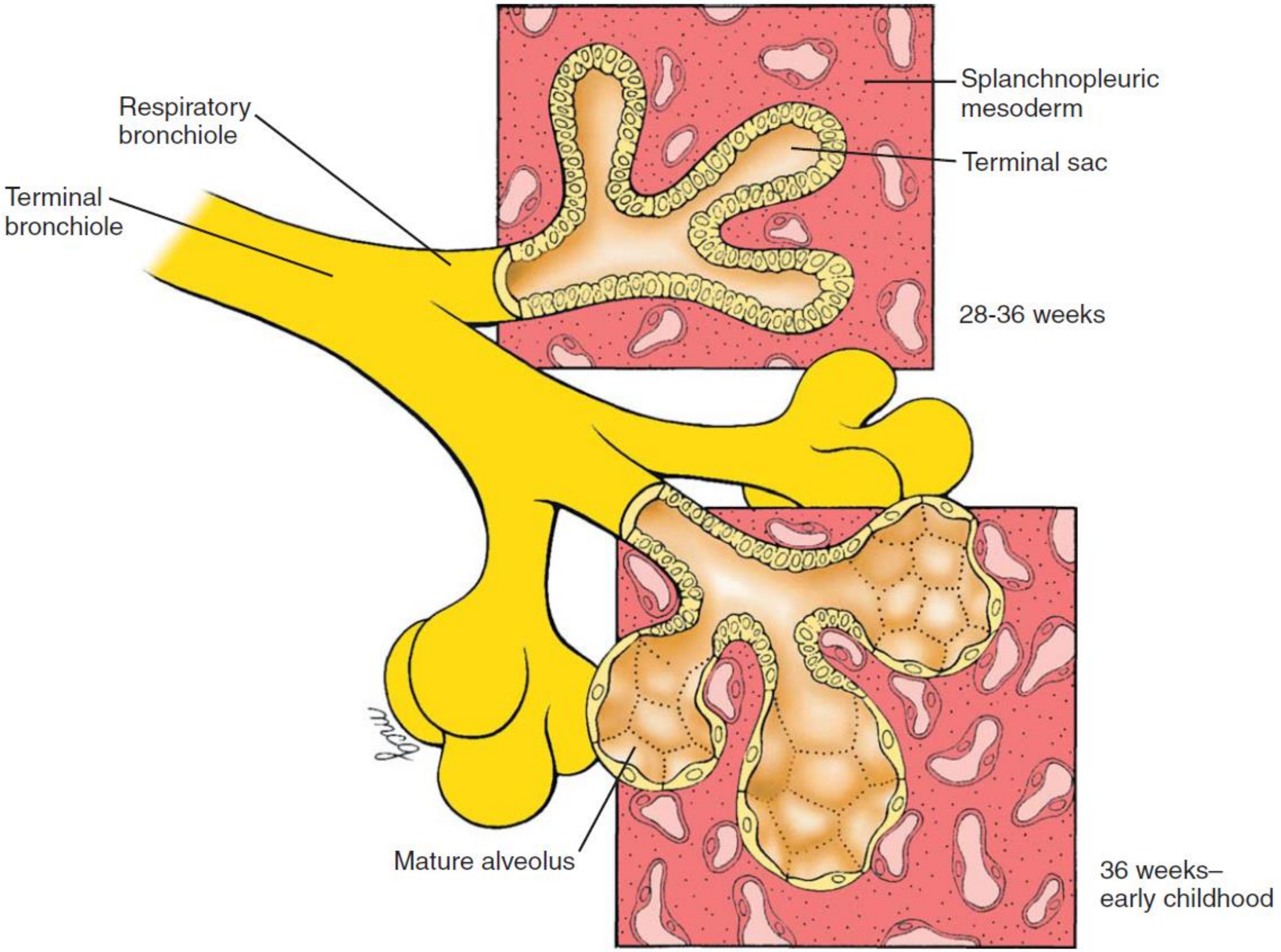
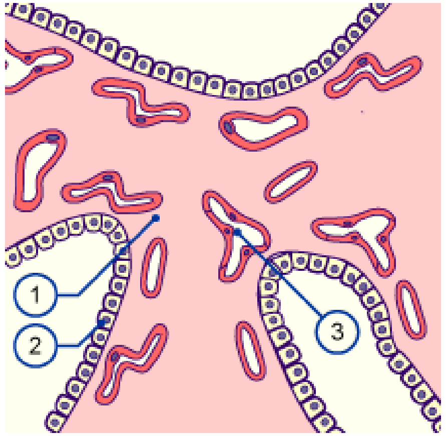


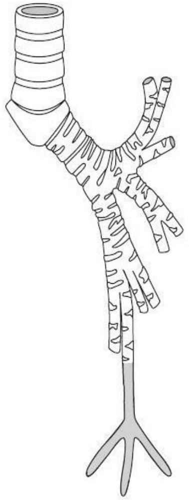
Figure 11-3. Maturation of lung tissue. Terminal sacs (primitive alveoli) begin to form between weeks 28 and 36 and begin to mature between 36 weeks and birth. However, only 5% to 20% of all terminal sacs eventually produced are formed before birth. Subsequent septation of the alveoli is not shown.

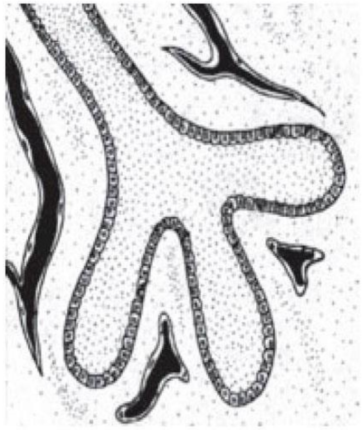
Table 11-1 Stages of Human Lung Development

Stage of Development	Period	Events
Embryonic	26 days to 6 weeks	Respiratory diverticulum arises as a ventral outpouching of foregut endoderm and undergoes three initial rounds of branching, producing the primordia successively of the two lungs, the lung lobes, and the bronchopulmonary segments; the stem of the diverticulum forms the trachea and larynx.
Pseudoglandular	6 to 16 weeks	Respiratory tree undergoes 14 more generations of branching, resulting in the formation of terminal bronchioles.
Canalicular	16 to 28 weeks	Each terminal bronchiole divides into two or more respiratory bronchioles. Respiratory vasculature begins to develop. During this process, blood vessels come into close apposition with the lung epithelium. The lung epithelium also begins to differentiate into specialized cell types (ciliated, secretory, and neuroendocrine cells proximally and precursors of the alveolar type II and I cells distally).
Saccular	28 to 36 weeks	Respiratory bronchioles subdivide to produce terminal sacs (primitive alveoli). Terminal sacs continue to be produced until well into childhood.
Alveolar	36 weeks to term	Alveoli mature.

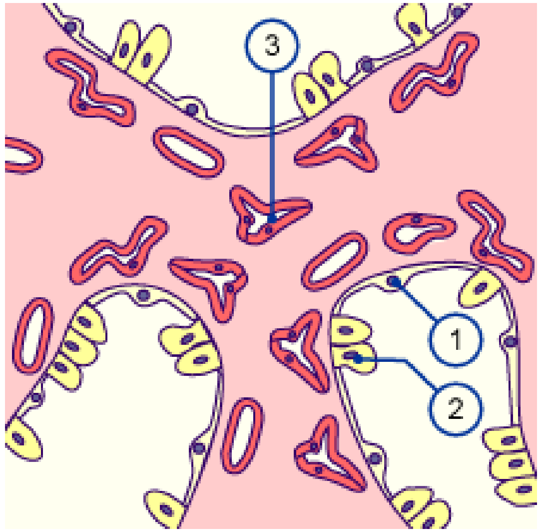
1. pseudoglandular period

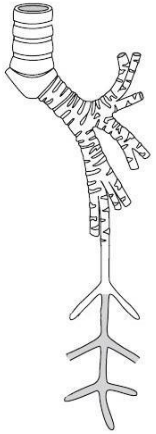


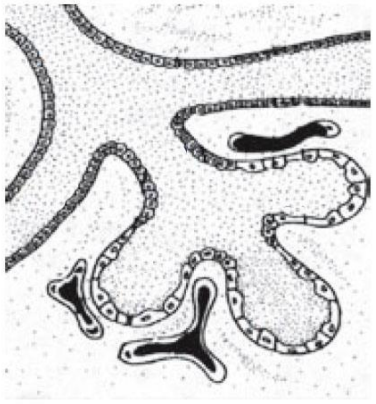




2. canalicular period







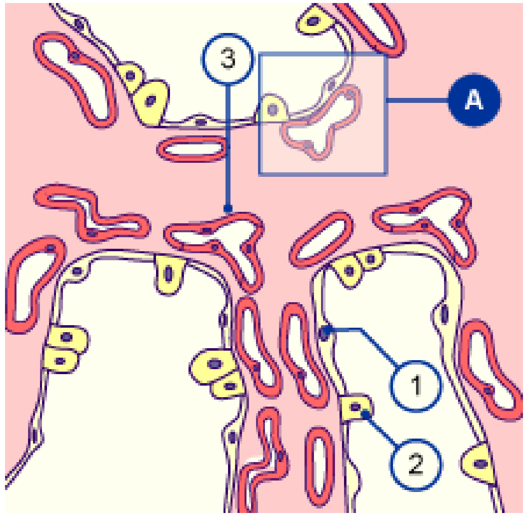
1. lung mesenchyme
2. type II. pneumocyte
3. capillary

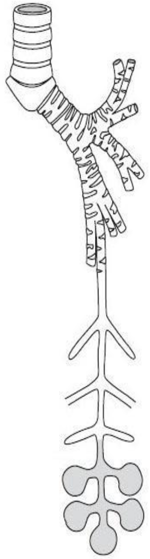
By week 16, after about 14 more branchings, the respiratory tree produces small branches called **terminal bronchioles.**

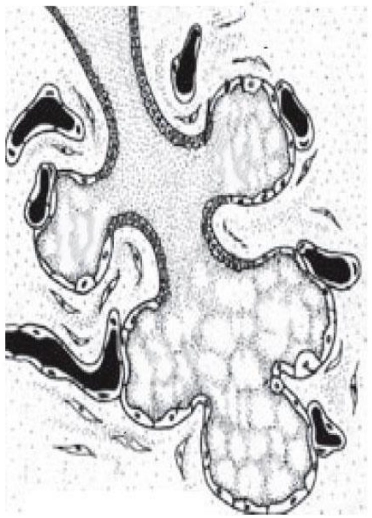
Type I. pneumocytes (1) differentiate from type II pneumocytes (2). Capillaries (3) grow towards the wall of the acini

Between 16 and 28 weeks each terminal bronchiole divides into two or more **respiratory bronchioles**, and the mesodermal tissue surrounding these structures **becomes highly vascularized.**

3. saccular period



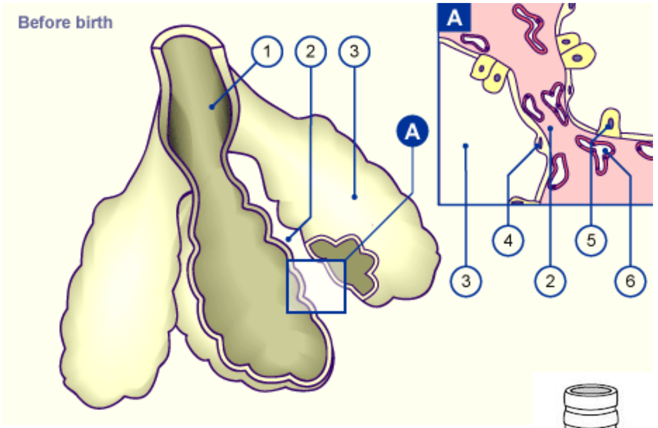




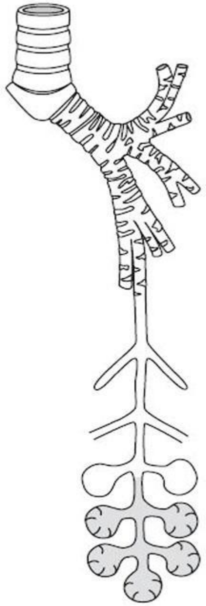
The number of capillaries increases around the acini. They push below the surface and begin to form a basal membrane common with the epithelium.

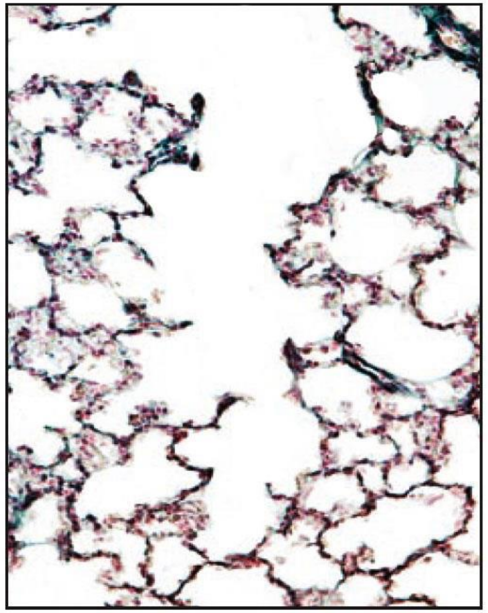
Respiratory bronchioles subdivide and produce terminal sac (primitive alveoli)

4. alveolar period: before birth



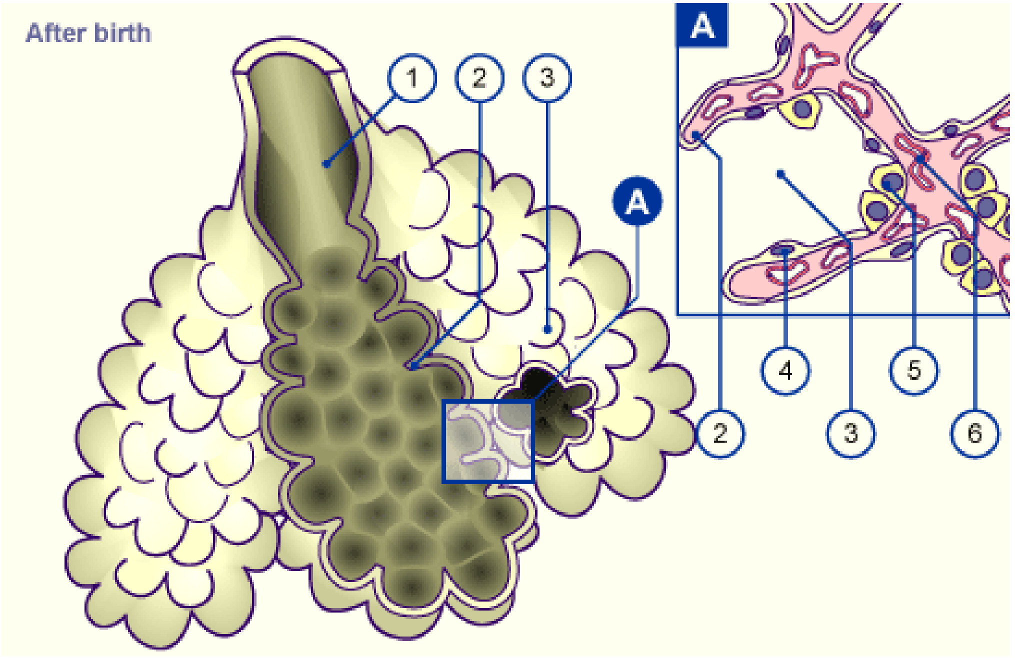
1. **Alveolar duct**
2. **Primary septum**
3. **Alveolar sac**
4. **Type I. pneumocyte**
5. **Type II. pneumocyte**
6. **capillaries**





Limited gas exchange is possible at this point, but the alveoli are still so few and immature that infants born at this age may die of respiratory insufficiency without adequate therapy.

Alveolar period: after birth

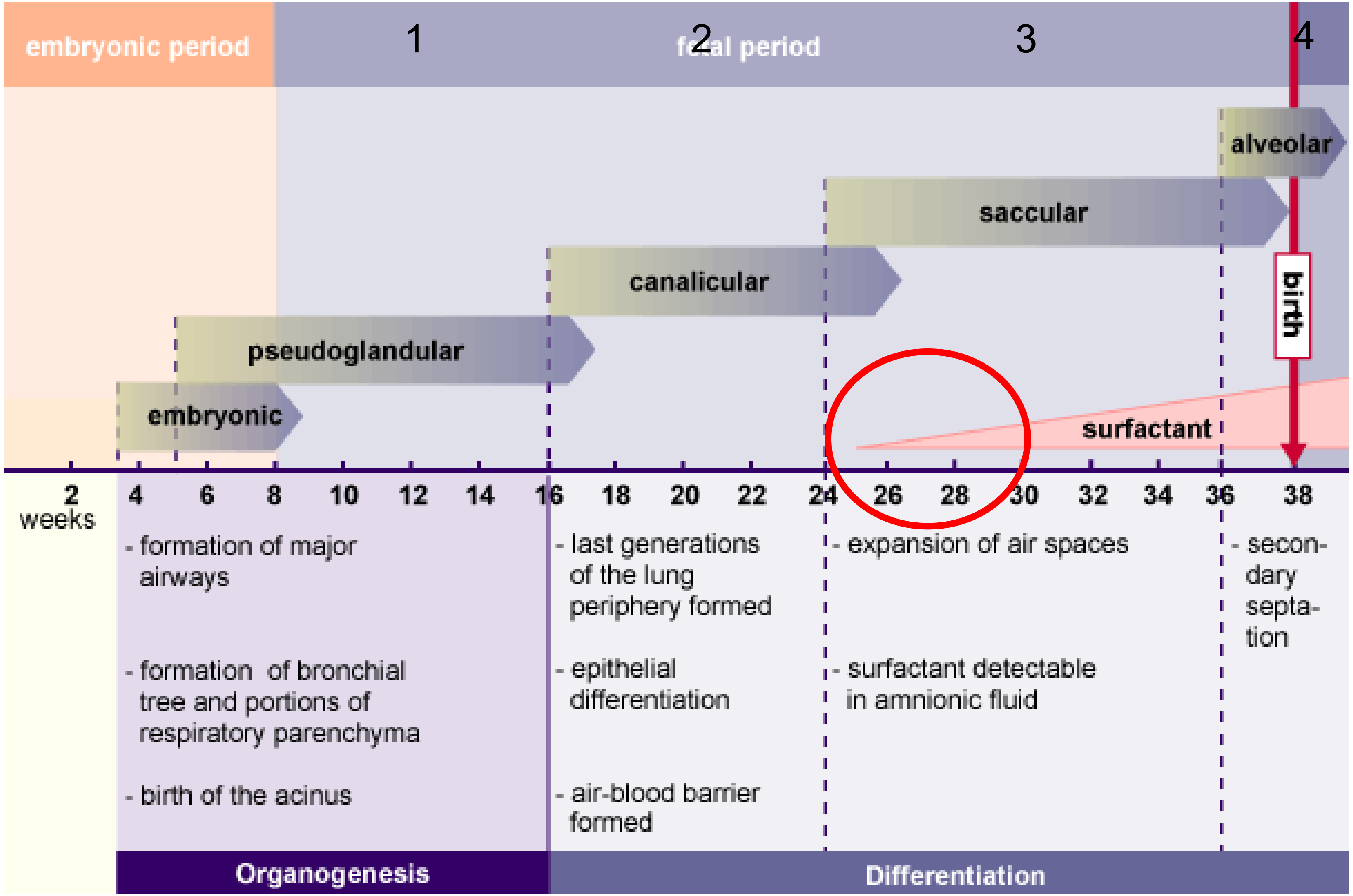


1. **Alveolar duct**
2. **Secondary septum**
3. **Alveolar sac**
4. **Type I. pneumocyte**
5. **Type II. pneumocyte**
6. **capillaries**

Additional terminal sacs continue to form and differentiate in craniocaudal progression both before and after birth. The process is largely completed by 2 years.

About 20 to 70 million terminal sacs are formed in each lung before birth; the total number of alveoli in the mature lung is 300 to 400 million. Continued thinning of the squamous epithelial lining of the terminal sacs begins just before birth, resulting in the differentiation of these primitive alveoli into mature alveoli. An important process of septation, which further subdivides the alveoli, occurs after birth. Each septum formed during this process contains smooth muscle and capillaries.

Without the surfactant layer the alveoli would collapse during expiration!
Respiratory distress syndrome, RDS, see next slide



Lung Maturation And Survival Of Premature Infants

As the end of gestation approaches, the lungs undergo a rapid and dramatic series of transformations that prepare them for air breathing. The fluid that fills the alveoli prenatally is absorbed at birth, the defenses that will protect the lungs against invading pathogens and against the oxidative effects of the atmosphere are activated, and the surface area for alveolar gas exchange increases greatly.

Changes in the structure of the lung take place during the last 2 months, accelerating in the days just preceding a normal term delivery. If a child is born prematurely, the state of development of the lungs is usually the prime factor determining whether he/she will live. Infants born between 26 weeks and term—during the phase of accelerated terminal lung maturation—have a very good chance of survival with appropriate neonatal support. Infants born before 26 weeks (in the canalicular phase of lung development) require intensive respiratory assistance to survive.

Even then, they may die or develop lung fibrosis that results in long-term respiratory problems.

Although the total surface area for gas exchange in the lung depends on the number of alveoli and on the density of alveolar capillaries, efficient gas exchange will occur only if the barrier separating air from blood is thin—that is, if the alveoli are thin-walled, properly inflated, and not filled with fluid. The walls of the maturing alveolar sacs thin out during the weeks before birth. In addition, specific alveolar cells (the alveolar type II cells) begin to secrete pulmonary surfactant, a mixture of phospholipids and surfactant proteins that reduces the surface tension of the liquid film lining the alveoli and thus facilitates inflation. In the absence of surfactant, the surface tension at the air-liquid interface of the alveolar sacs tends to collapse the alveoli during exhalation. These collapsed alveoli can be inflated only with great effort. The primary cause of the **respiratory distress syndrome of premature infants** (pulmonary insufficiency accompanied by gasping and cyanosis) is an inadequate production of surfactant.

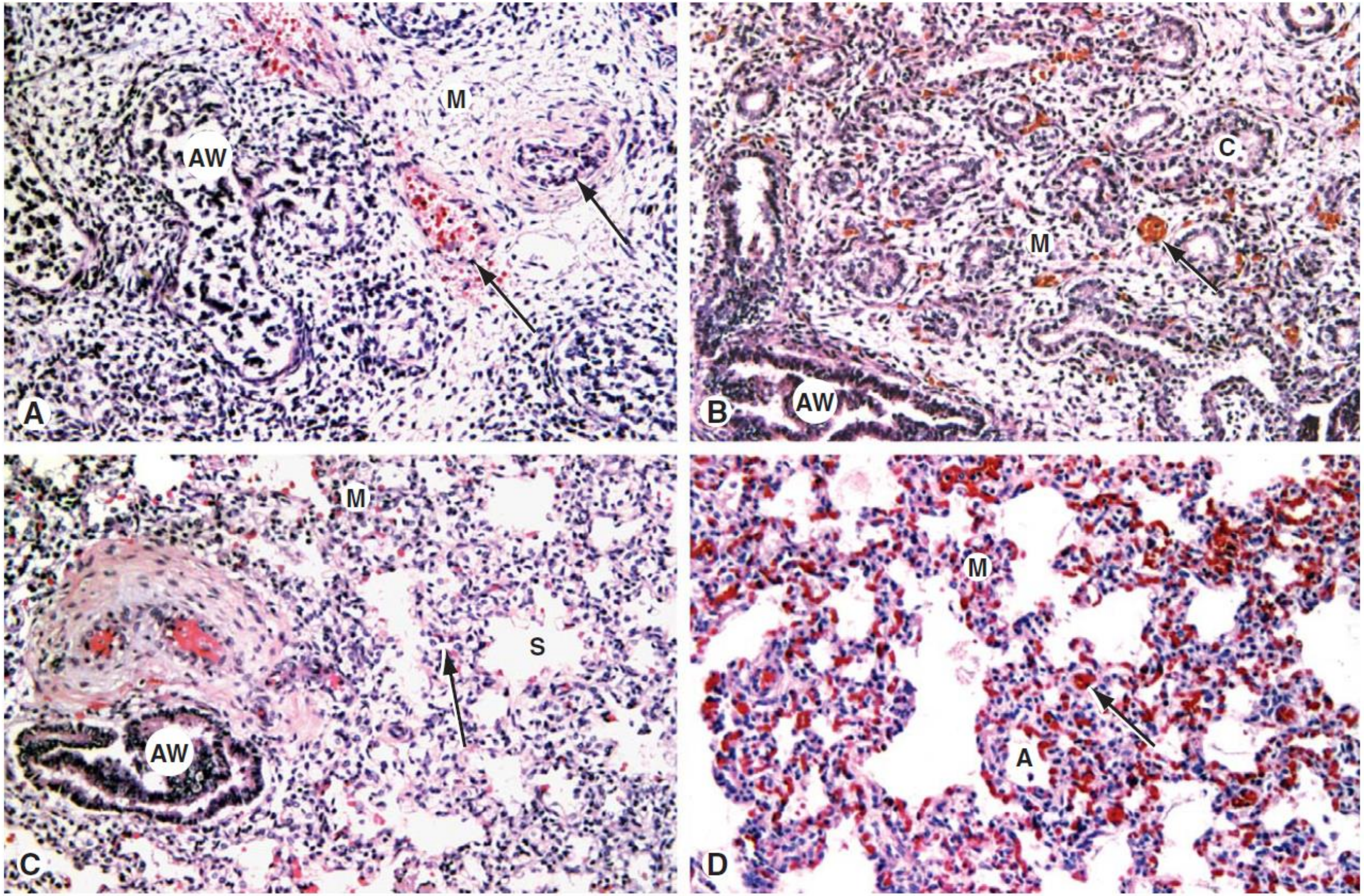
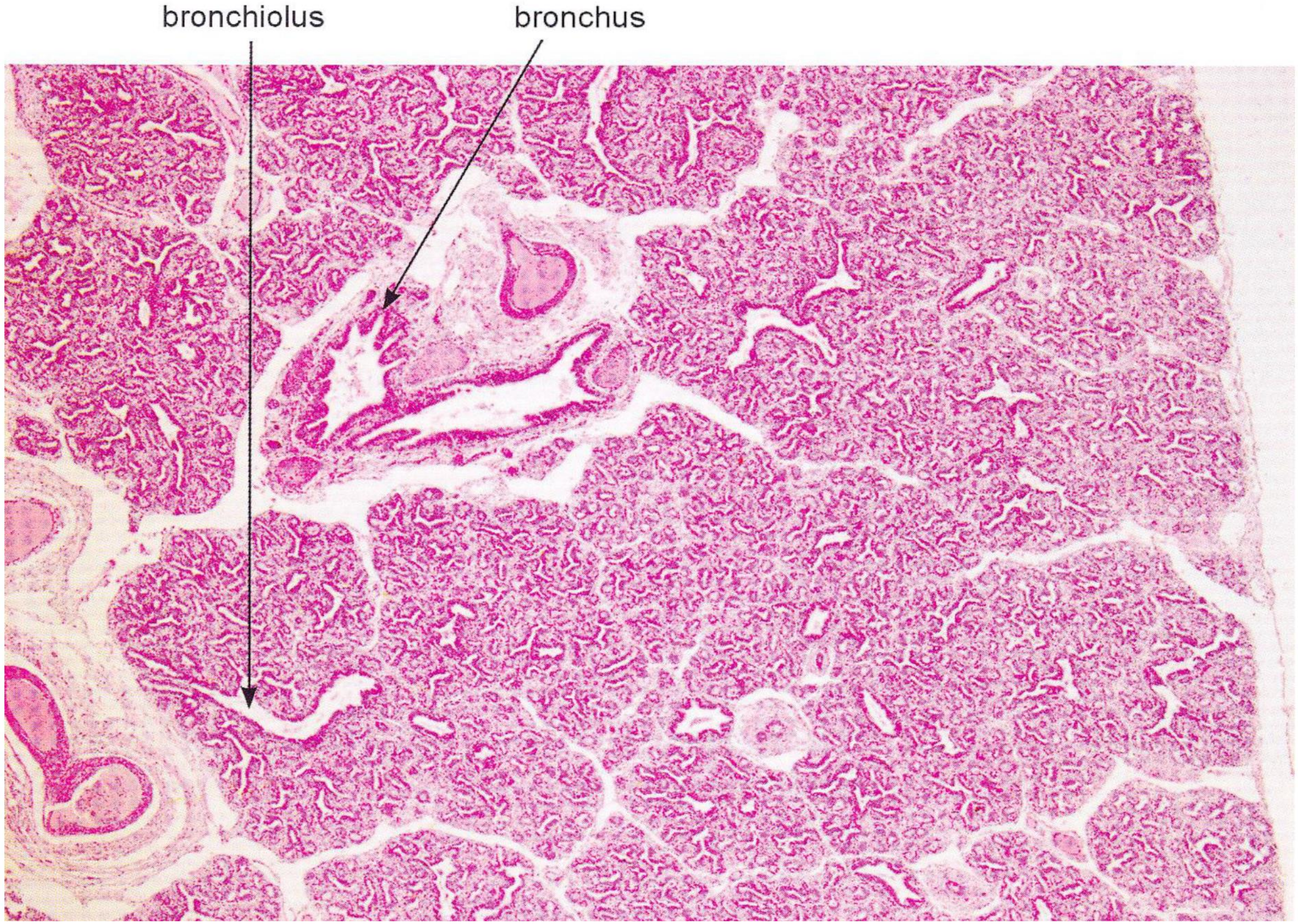


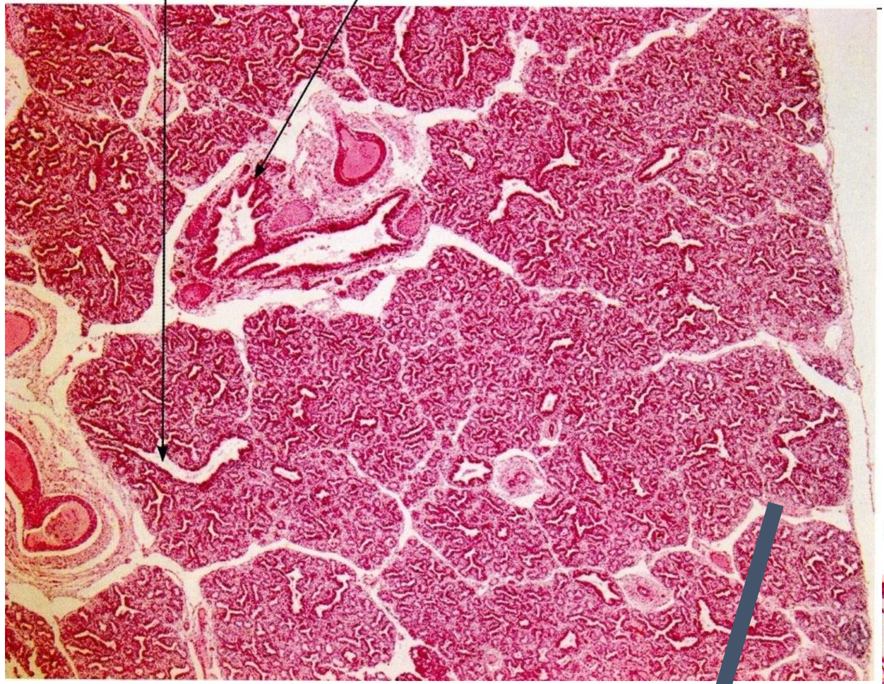
Figure 11-2. Histologic stages of human lung development. *A*, Pseudoglandular stage. *B*, Canalicular stage. *C*, Saccular stage. *D*, Alveolar stage. A, alveolus; AW, airway; C, smooth-walled canaliculus; M, mesenchyme; S, saccule; *Arrows*, part A, pulmonary artery (thick wall) and vein (thin wall); *Arrows*, parts B-D, capillaries.

Fetal lung section

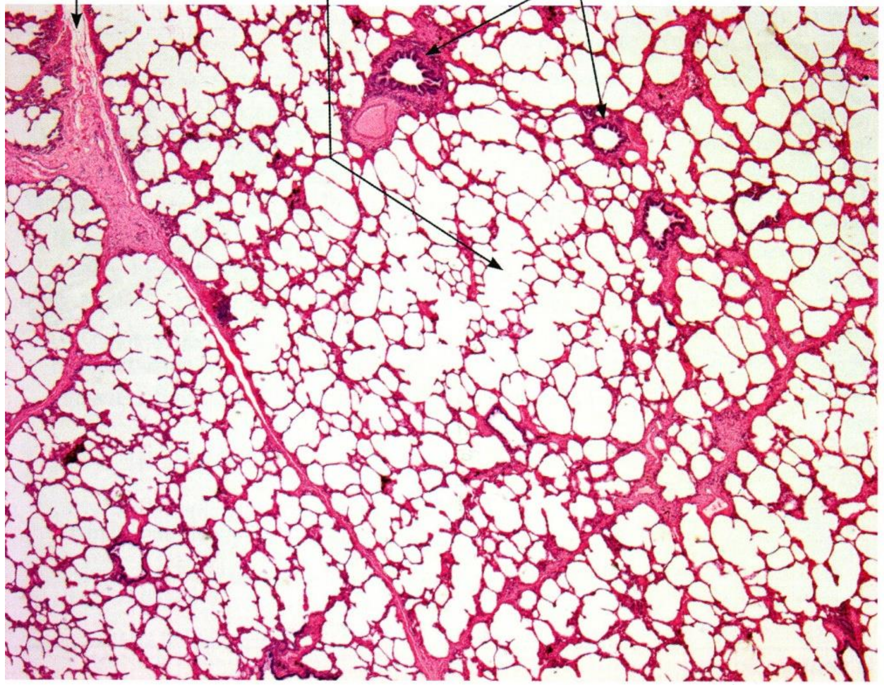
looks like a gland



Fetal lung section



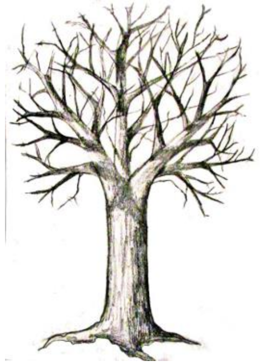
before birth



after birth

Recognition:
Dichotomic divisions
Hyalin cartilage

Development of the respiratory tree
A brief summary



On day 22 the **foregut** produces a ventral evagination called the **respiratory diverticulum** or lung bud, which is the primordium of the lungs. As the lung bud grows, it remains ensheathed in a covering of **splanchnopleuric mesoderm**, which will give rise to the lung vasculature and to the connective tissue, cartilage, and muscle within the bronchi.

On days 26 to 28 the lengthening lung bud bifurcates into ***left and right primary bronchial buds***, which will give rise to the two lungs.

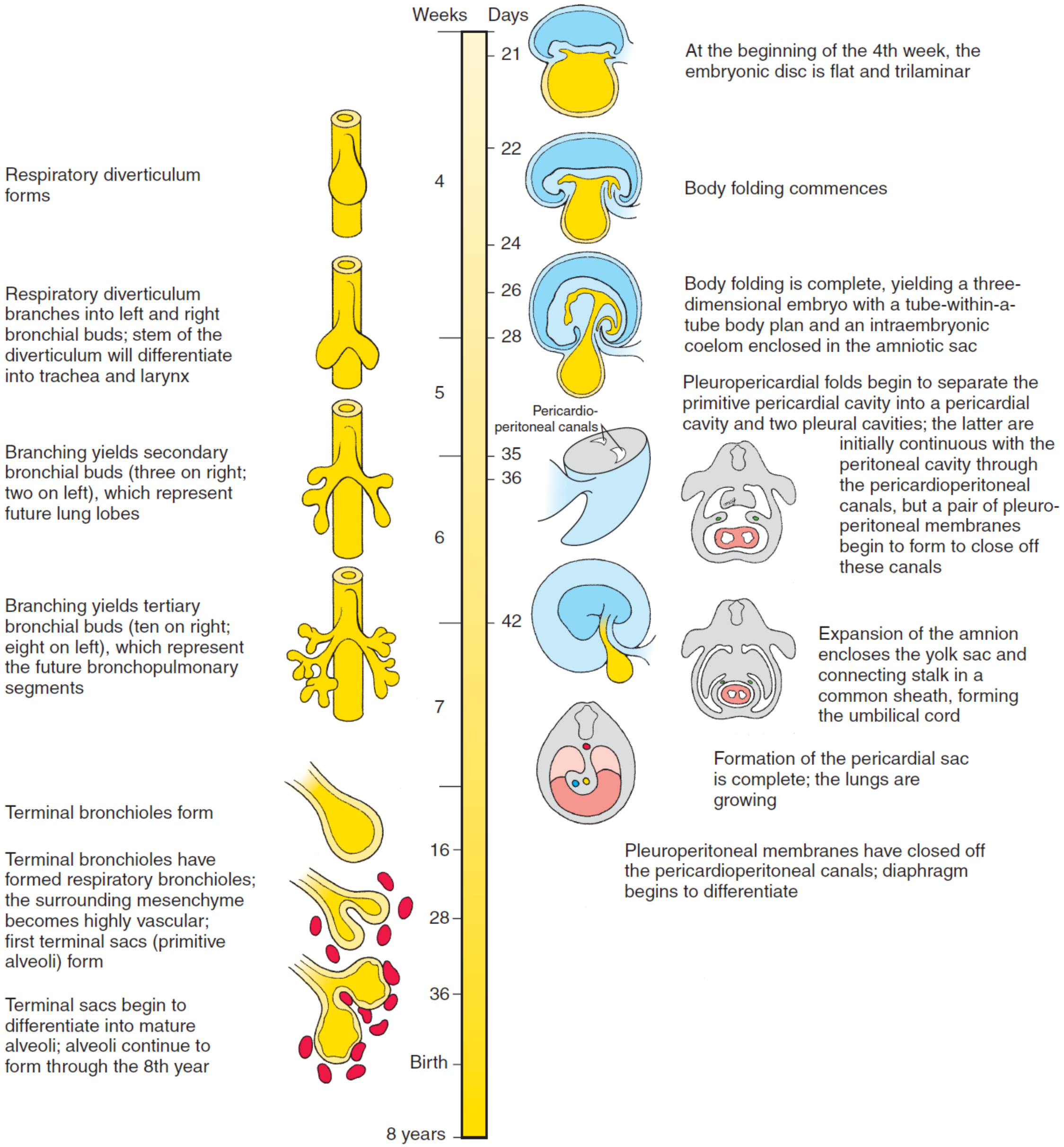
In the 5th week a second generation of branching produces three secondary bronchial buds on the right side and two on the left. These **are the primordia of the future lung lobes**. The bronchial buds and their splanchnopleuric sheath continue to grow and bifurcate, gradually filling the **pleural cavities**.

By week 28 the 16th round of branching generates **terminal bronchioles**, which subsequently divide into two or more **respiratory bronchioles**.

By week 36 these respiratory bronchioles have become invested with capillaries and are called **terminal sacs or primitive alveoli.**

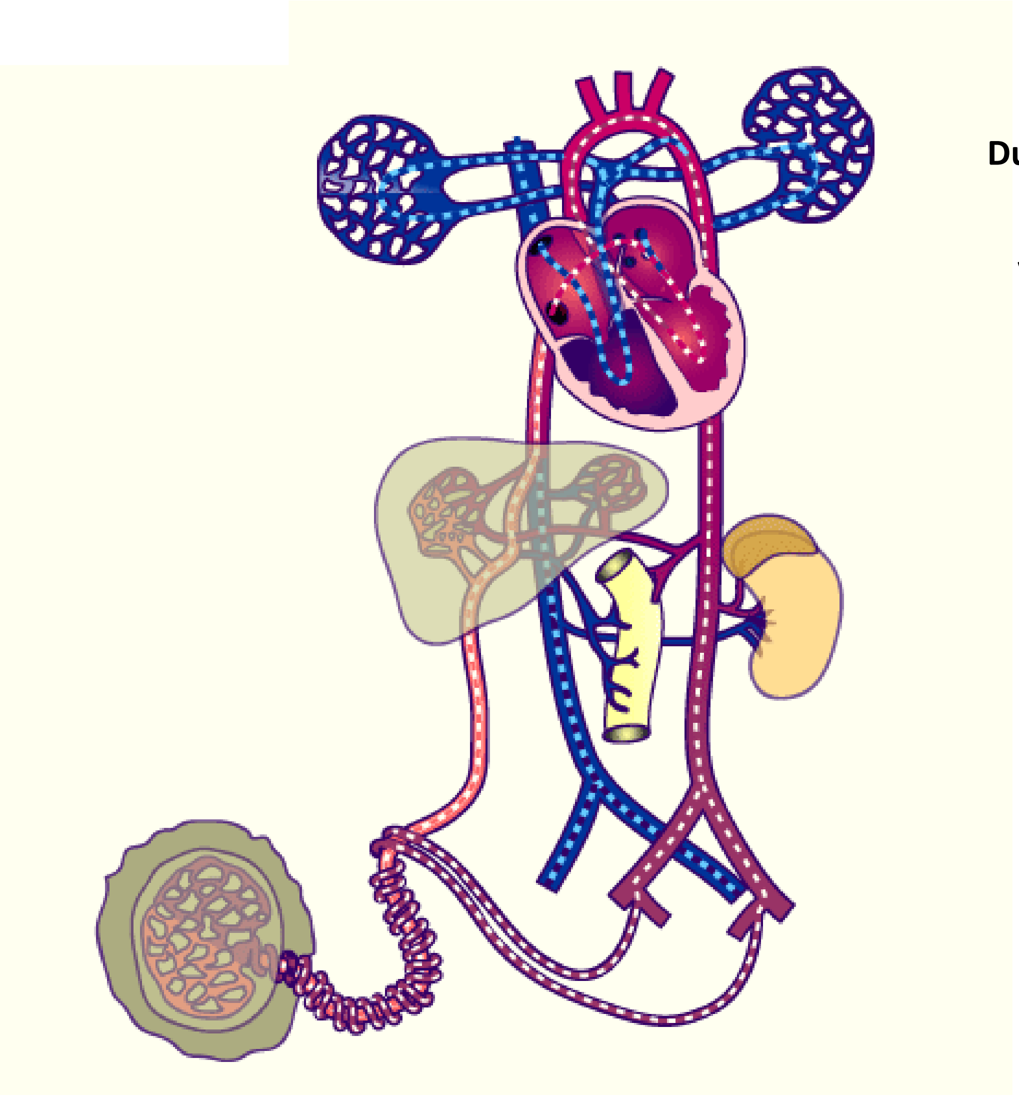
Between 36 weeks and birth, the alveoli mature. **Additional alveoli** continue to be produced **throughout early childhood.**

Time lines of respiratory development together with body cavity formation



Time line. Development of the lungs, respiratory tree, and body cavities.

Fetal circulation



Placenta → Umbilical vein → Ductus venosus → Vena cava inferior → Right atrium → Foramen ovale → Left atrium → Left ventricle → Aorta → Umbilical artery → Placenta

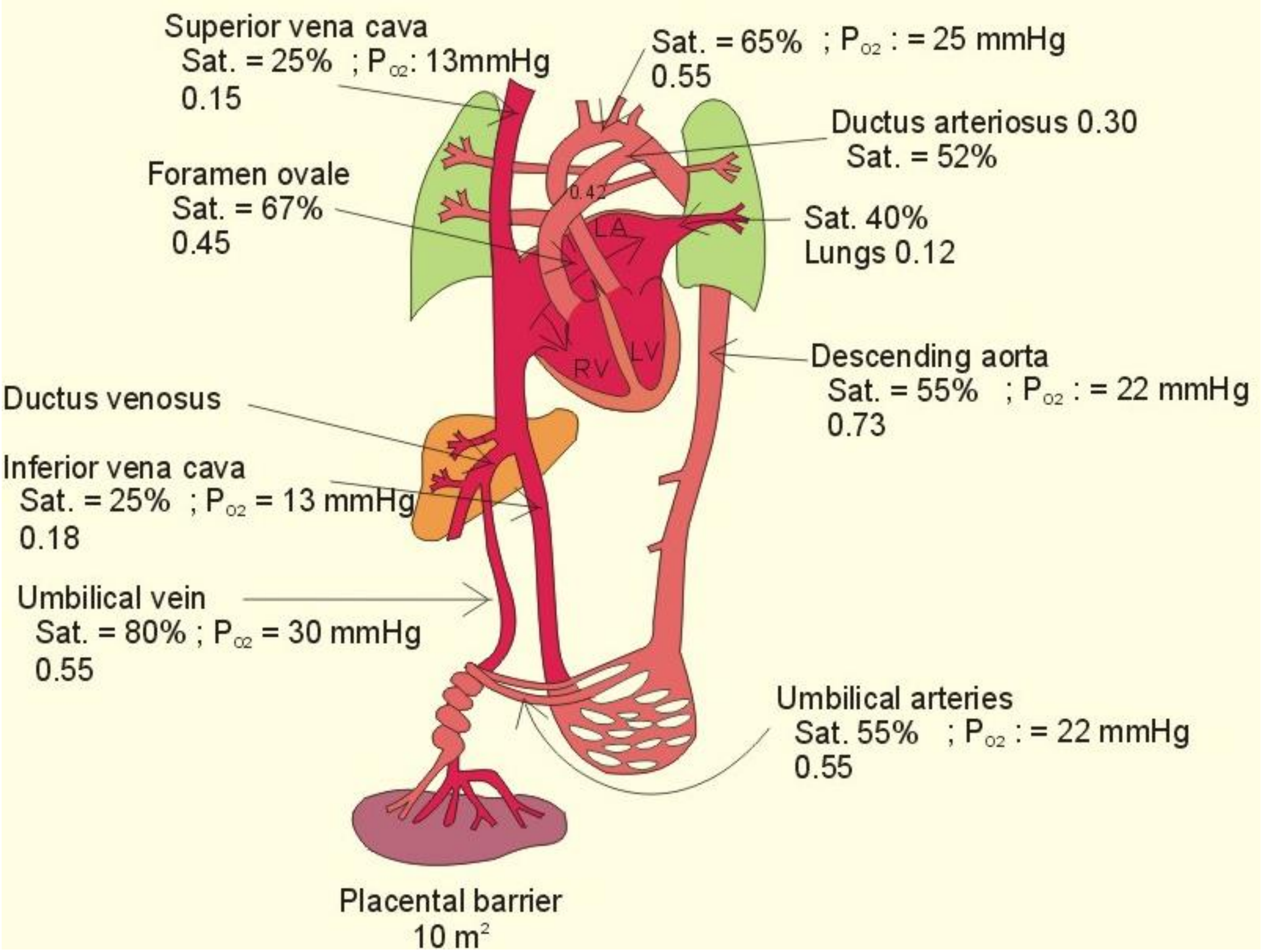
Umbilical vein → Liver sinusoids → Vena cava inferior

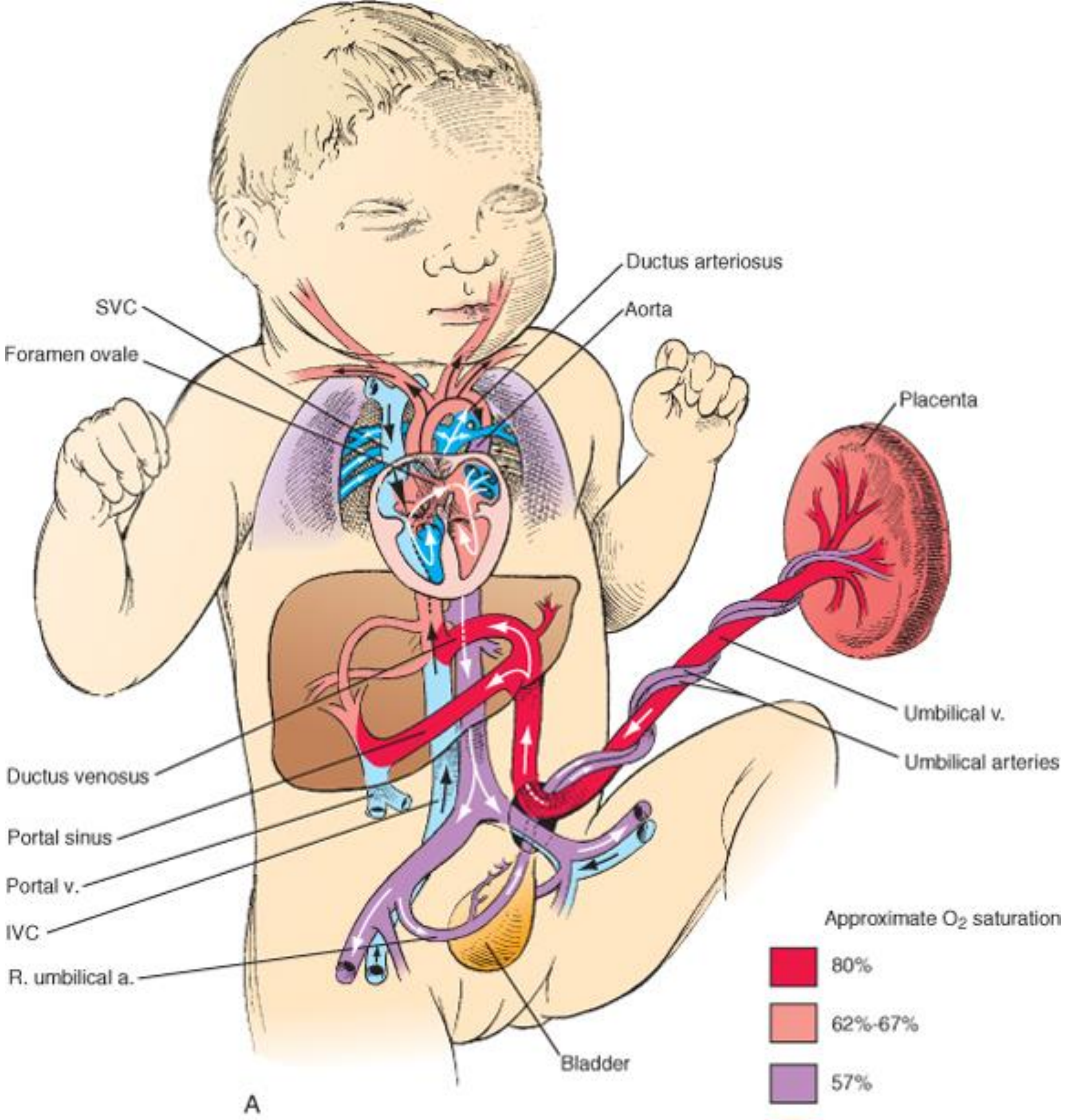
Superior vena cava → Right atrium

Right atrium → Right ventricle → Truncus pulmonalis → Lungs → Venae pulmonales → Left atrium

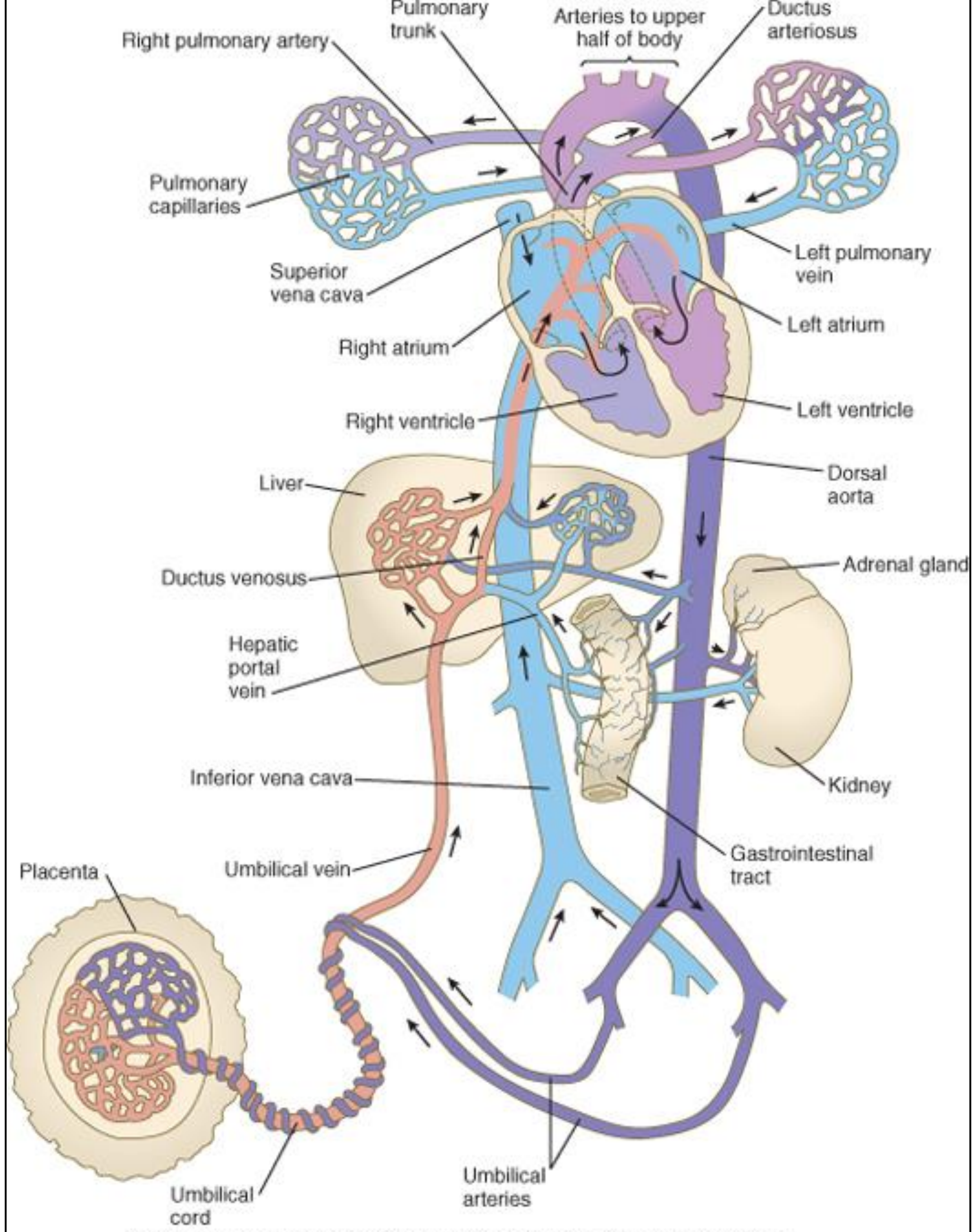
Truncus pulmonalis → Ductus arteriosus → Aorta

Oxygen saturation of the fetal blood





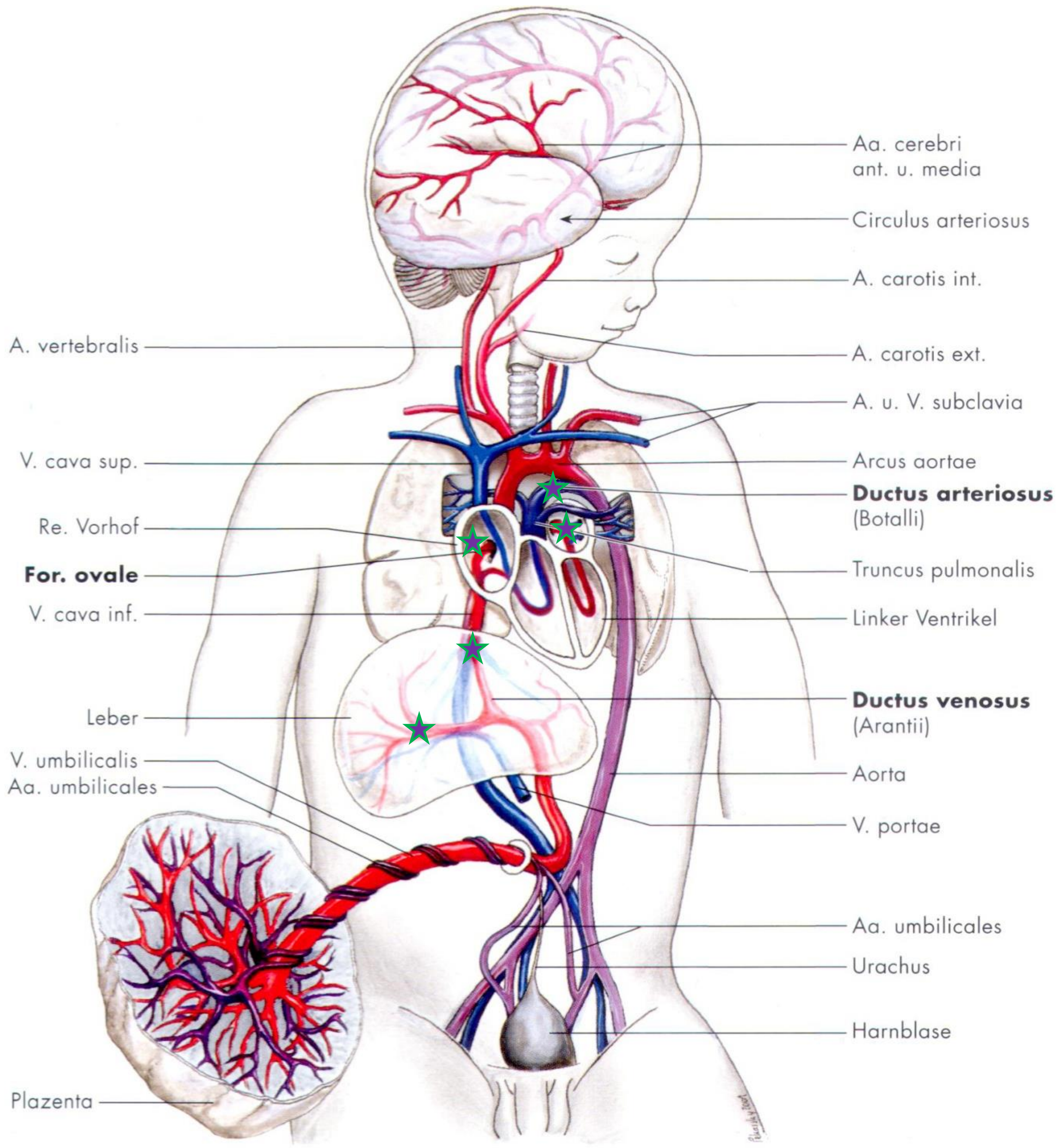
Schoenwolf et al: Larsen's Human Embryology, 4th Edition.



Carlson: Human Embryology and Developmental Biology, 4th Edition.

Fetal circulation at term

Right-left shunts in the fetal circulation



Liver sinusoids

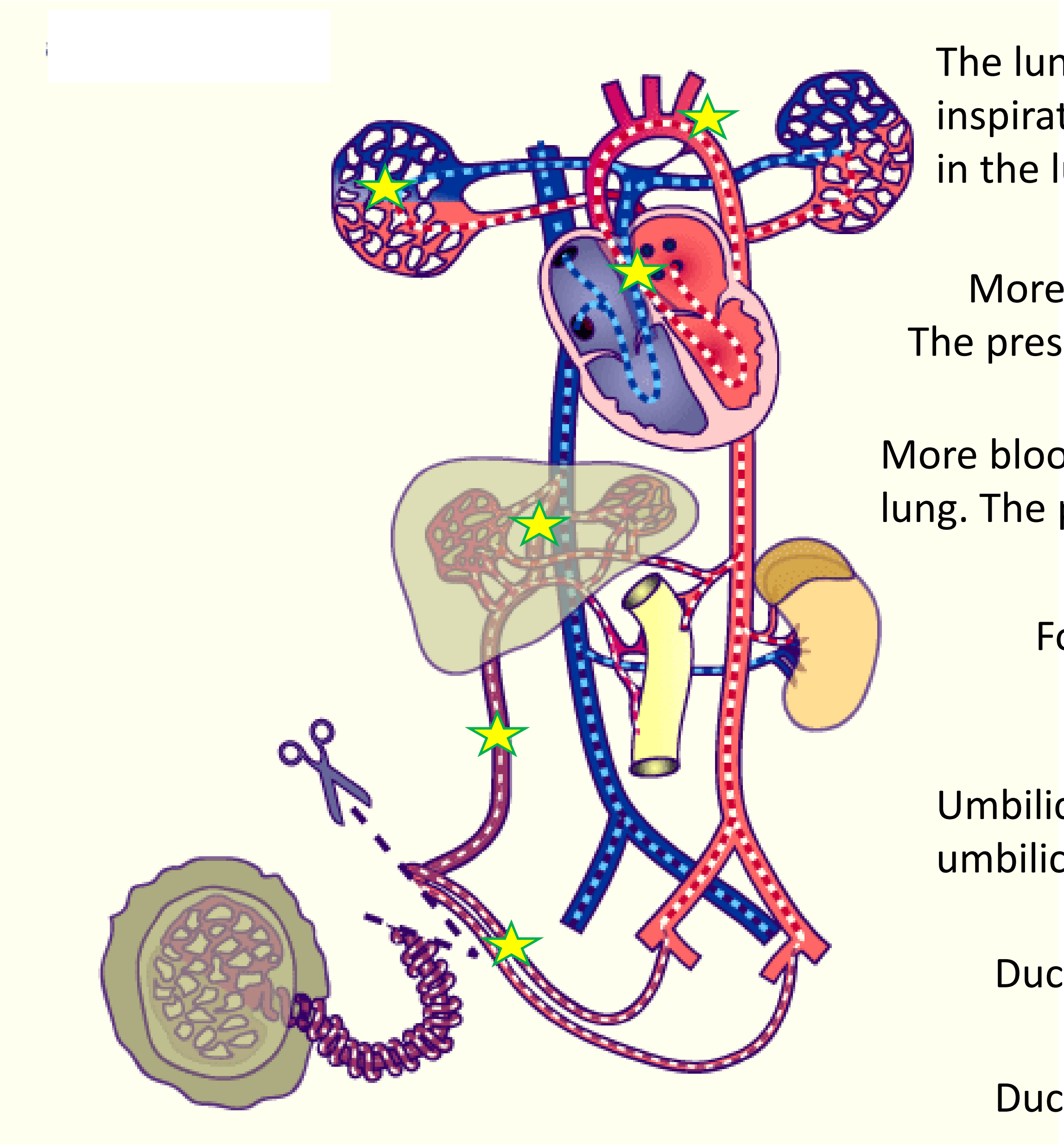
Vena umbilicalis entrance to the vena cava inferior

Right atrium
(Vena cava superior)

Left atrium
(Venae pulmonales)

Ductus arteriosus
entrance to the aortic arch

Perinatal changes of the circulation



The lung alveoli open with the first inspiration. The lung expands. The resistance in the lung vessels decreases.

⬇

More blood flows toward the lungs.
The pressure in the right atrium decreases.

⬇

More blood flows to the left atrium from the lung. The pressure on the left atrium increases.

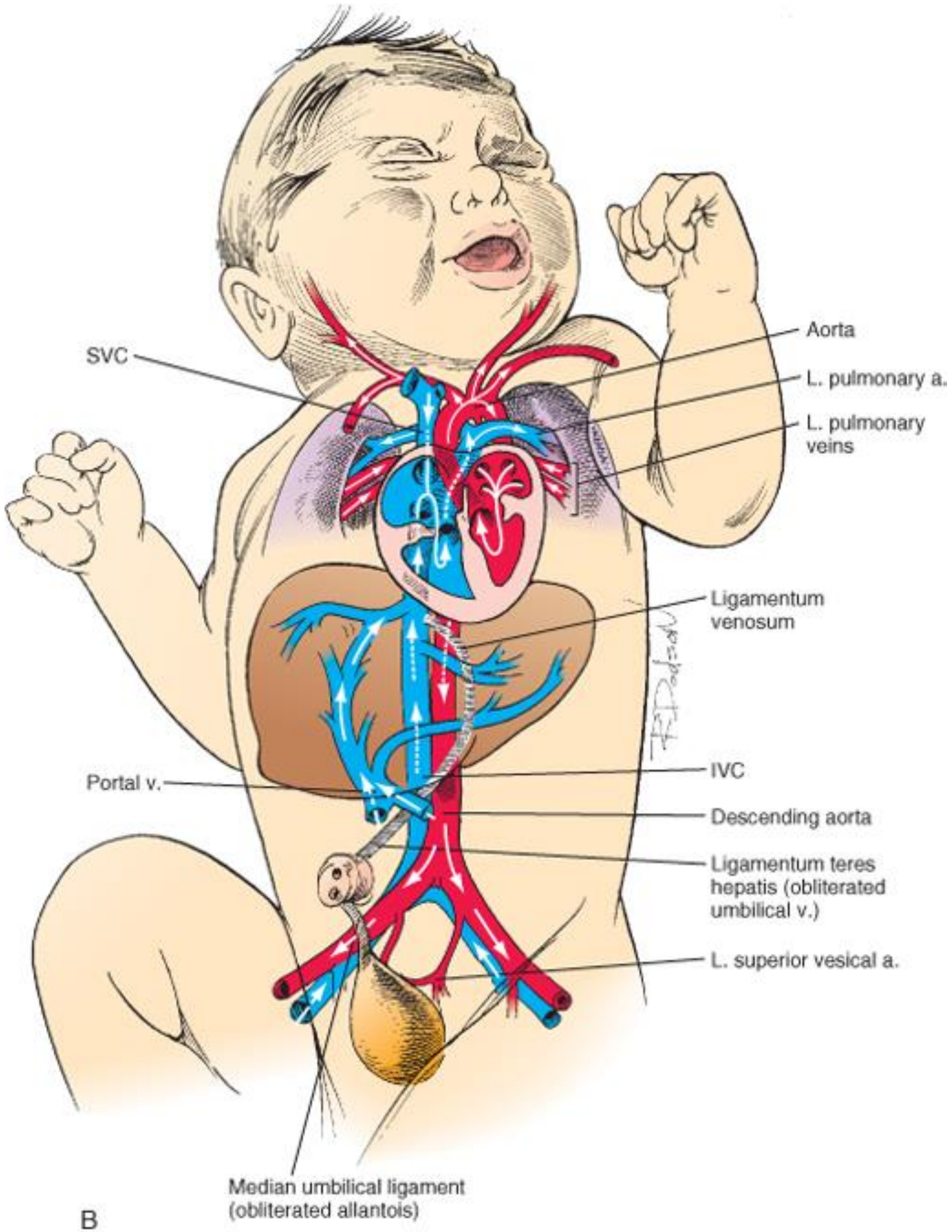
⬇

Foramen ovale is closed.

Umbilical vein and the distal part of the umbilical artery is obliterated.

Ductus venosus is obliterated.

Ductus arteriosus is obliterated.



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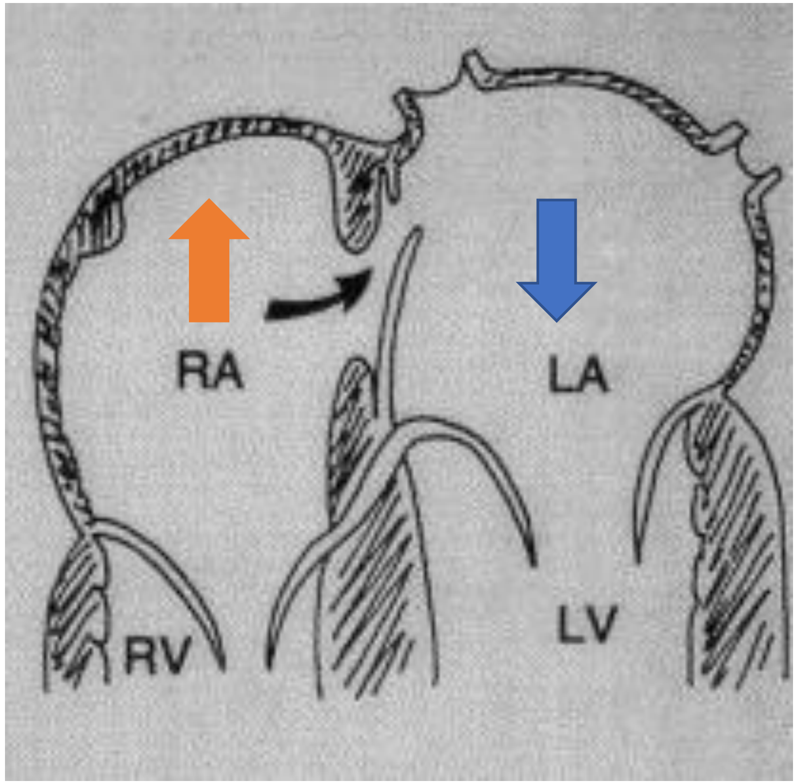
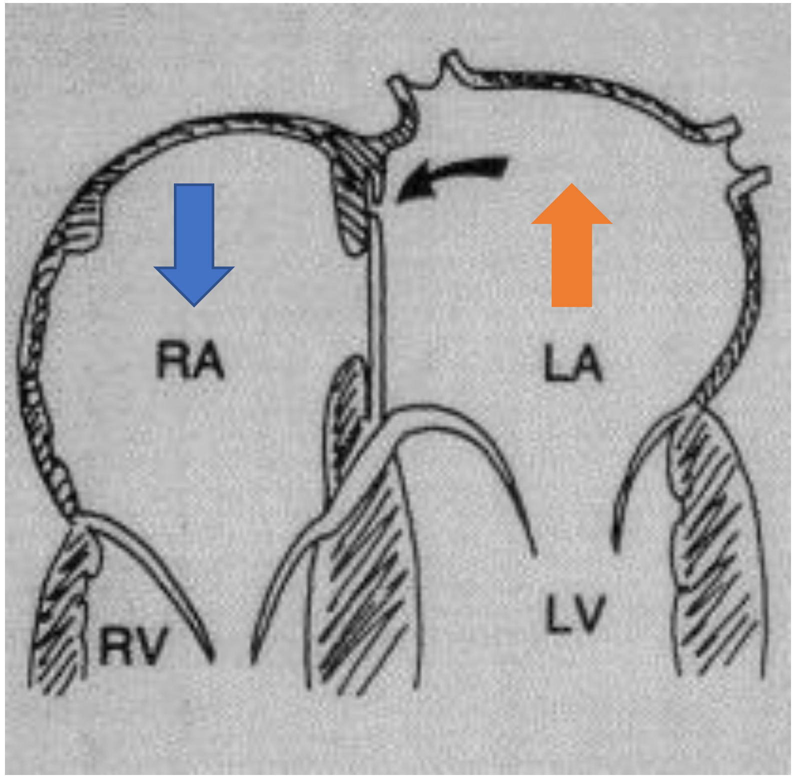
Closure of the foramen ovale

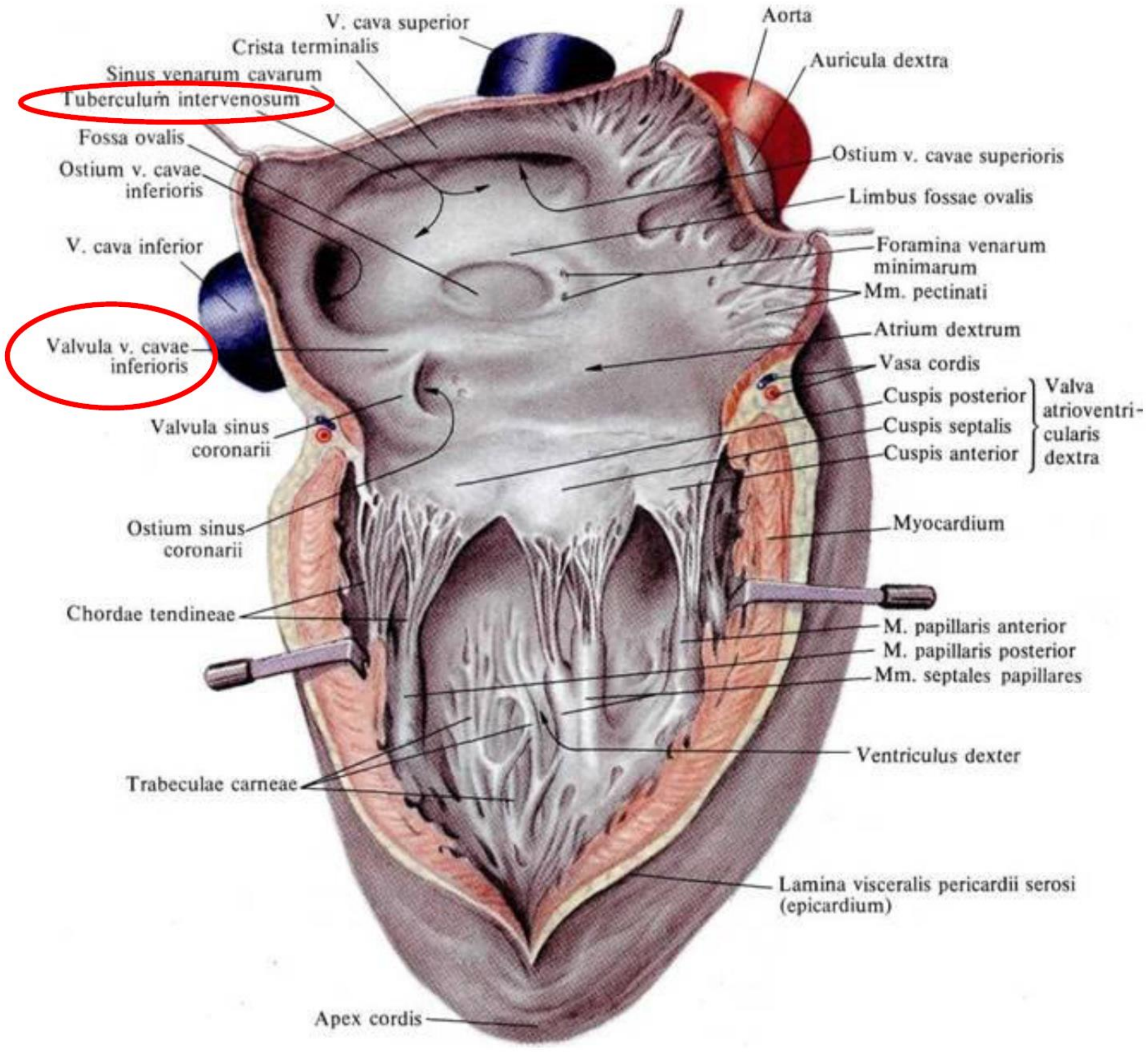


before birth



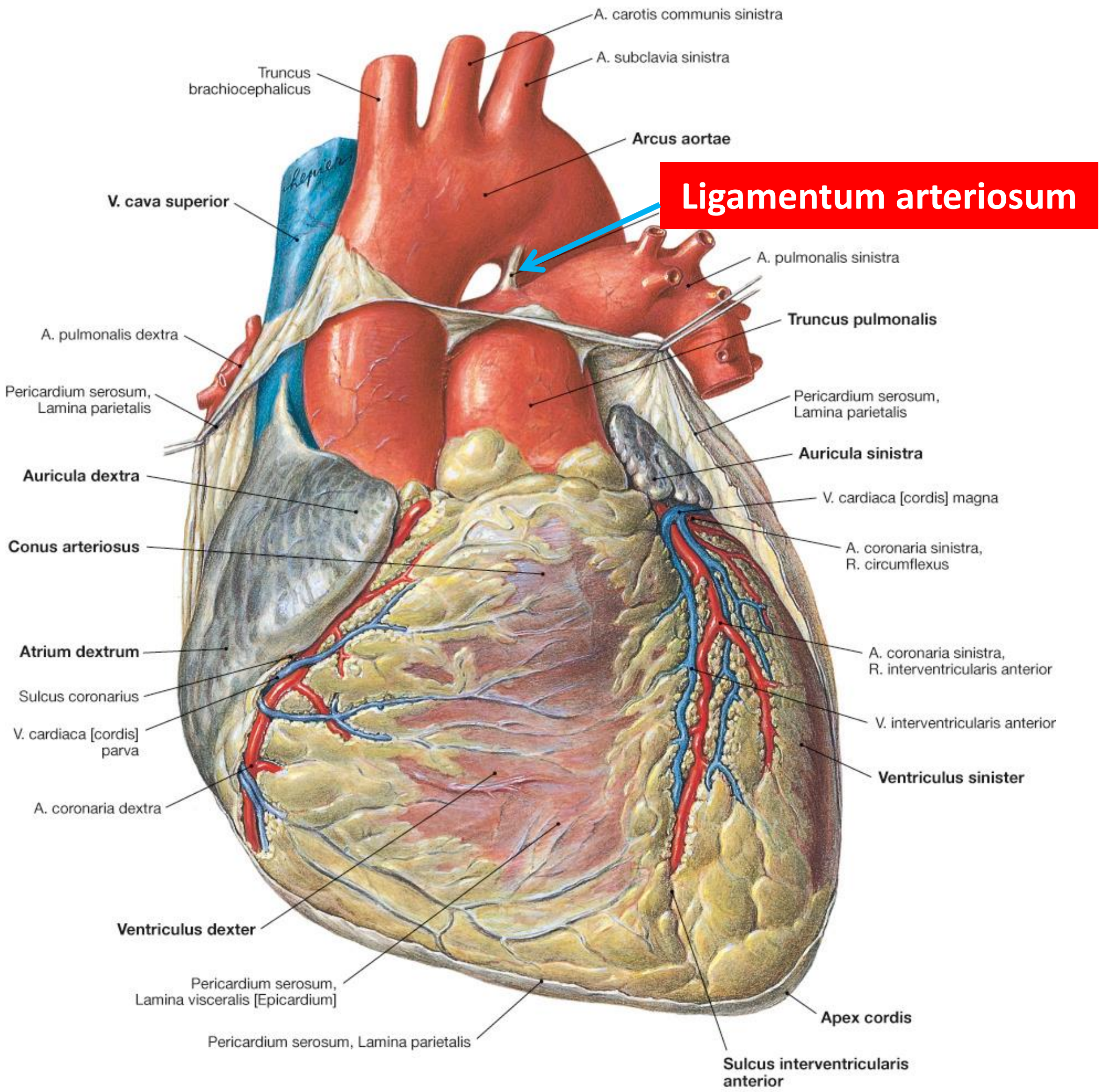
after birth

David R Holmes Jr et al. Current Problems in Cardiology, 2004,29:56-94

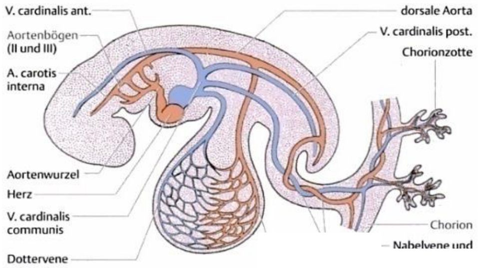


V. cava superior
Crista terminalis
Sinus venarum cavarum
Tuberculum intervenosum
Fossa ovalis
Ostium v. cavae inferioris
V. cava inferior
Valvula v. cavae inferioris
Valvula sinus coronarii
Ostium sinus coronarii
Chordae tendineae
Trabeculae carneae
Apex cordis
Aorta
Auricula dextra
Ostium v. cavae superioris
Limbus fossae ovalis
Foramina venarum minimarum
Mm. pectinati
Atrium dextrum
Vasa cordis
Cuspis posterior
Cuspis septalis
Cuspis anterior
Valva atrioventri-cularis dextra
Myocardium
M. papillaris anterior
M. papillaris posterior
Mm. septales papillares
Ventriculus dexter
Lamina visceralis pericardii serosi (epicardium)

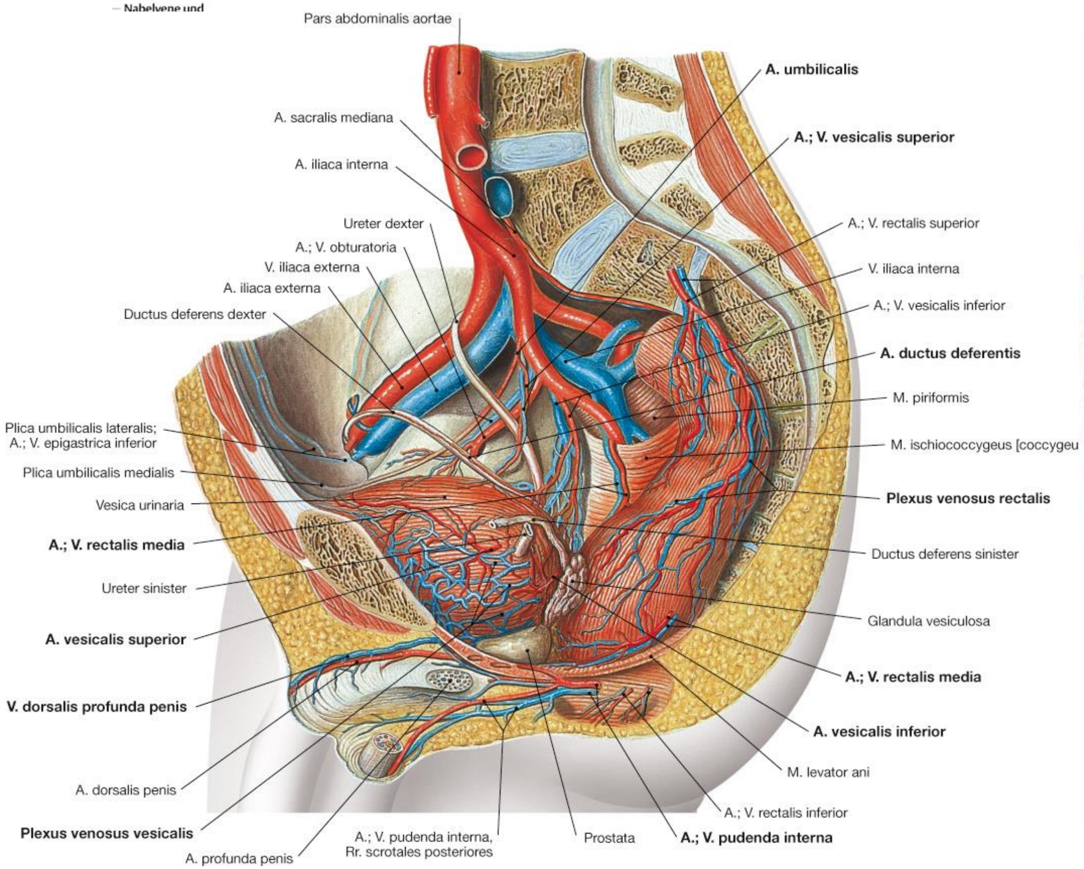
Remnants

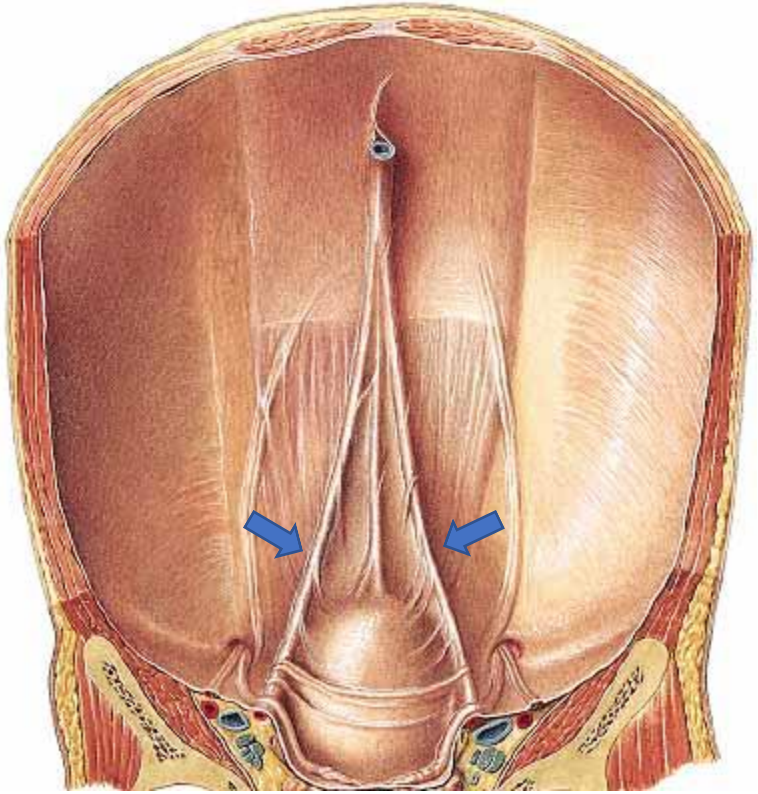


Ligamentum arteriosum
A. carotis communis sinistra
A. subclavia sinistra
Truncus brachiocephalicus
Arcus aortae
V. cava superior
A. pulmonalis sinistra
Truncus pulmonalis
A. pulmonalis dextra
Pericardium serosum, Lamina parietalis
Pericardium serosum, Lamina parietalis
Auricula sinistra
Auricula dextra
V. cardiaca [cordis] magna
Conus arteriosus
A. coronaria sinistra, R. circumflexus
Atrium dextrum
A. coronaria sinistra, R. interventricularis anterior
Sulcus coronarius
V. interventricularis anterior
V. cardiaca [cordis] parva
Ventriculus sinister
A. coronaria dextra
Ventriculus dexter
Pericardium serosum, Lamina visceralis [Epicardium]
Apex cordis
Pericardium serosum, Lamina parietalis
Sulcus interventricularis anterior

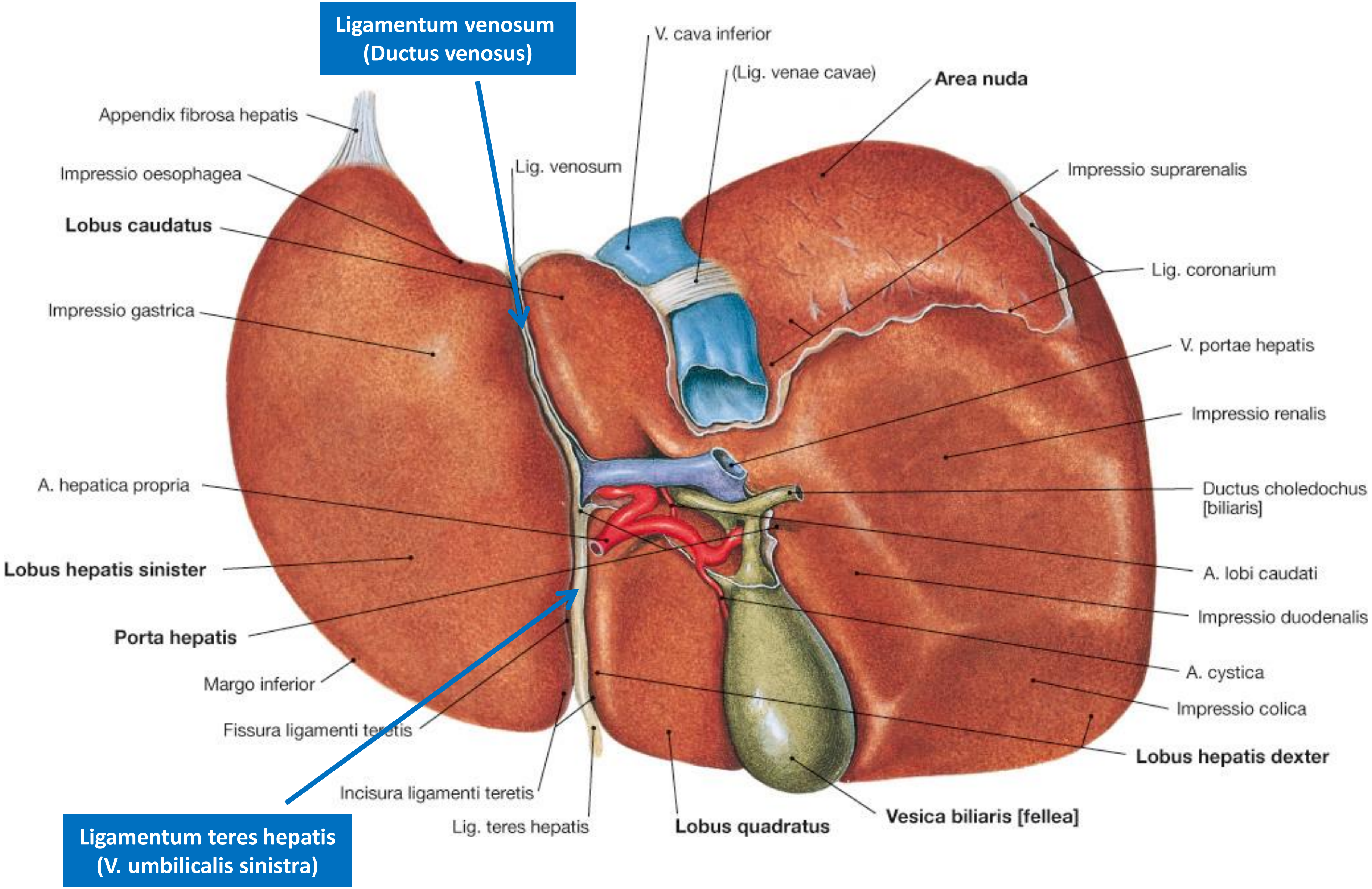

V. cardinalis ant.
Aortenbögen (II und III)
A. carotis interna
Aortenwurzel
Herz
V. cardinalis communis
Dottervene
dorsale Aorta
V. cardinalis post.
Chorionzotte
Chorion
Nabelvene und

Remnants


Pars abdominalis aortae
A. sacralis mediana
A. iliaca interna
Ureter dexter
A.; V. obturatoria
V. iliaca externa
A. iliaca externa
Ductus deferens dexter
Plica umbilicalis lateralis; A.; V. epigastrica inferior
Plica umbilicalis medialis
Vesica urinaria
A.; V. rectalis media
Ureter sinister
A. vesicalis superior
V. dorsalis profunda penis
A. dorsalis penis
Plexus venosus vesicalis
A. profunda penis
A.; V. pudenda interna, Rr. scrotales posteriores
Prostata
A.; V. pudenda interna
A.; V. rectalis inferior
M. levator ani
A. vesicalis inferior
A.; V. rectalis media
Glandula vesiculosa
Ductus deferens sinister
Plexus venosus rectalis
M. ischiococcygeus [coccygeu
M. piriformis
A. ductus deferentis
A.; V. vesicalis inferior
V. iliaca interna
A.; V. rectalis superior
A.; V. vesicalis superior
A. umbilicalis

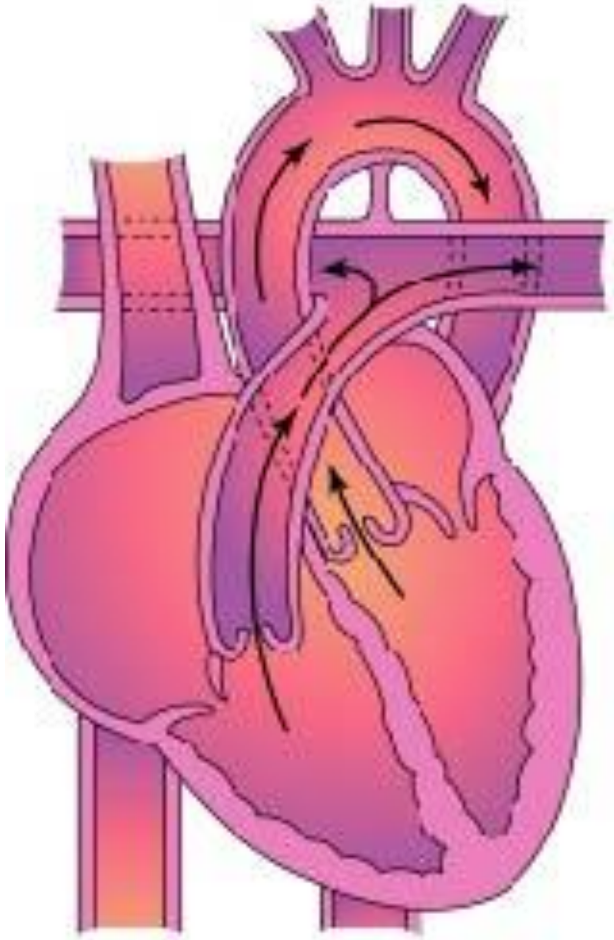


Remnants

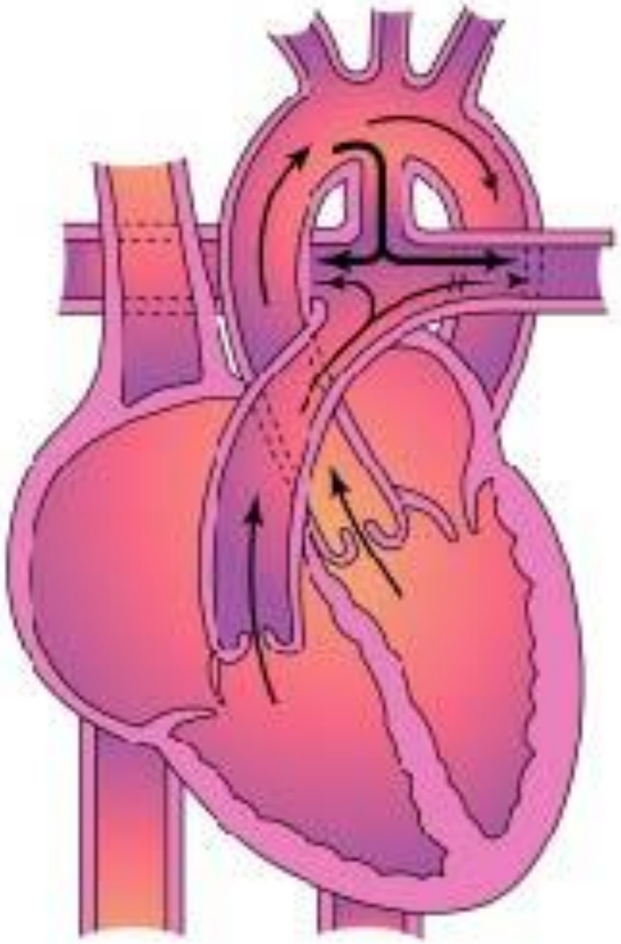


Ligamentum venosum
(Ductus venosus)
V. cava inferior
(Lig. venae cavae)
Area nuda
Appendix fibrosa hepatis
Lig. venosum
Impressio oesophagea
Impressio suprarenalis
Lobus caudatus
Lig. coronarium
Impressio gastrica
V. portae hepatis
Impressio renalis
A. hepatica propria
Ductus choledochus
[biliaris]
Lobus hepatis sinister
A. lobi caudati
Impressio duodenalis
Porta hepatis
A. cystica
Margo inferior
Impressio colica
Fissura ligamenti teretis
Lobus hepatis dexter
Incisura ligamenti teretis
Lig. teres hepatis
Lobus quadratus
Vesica biliaris [fellea]
Ligamentum teres hepatis
(V. umbilicalis sinistra)

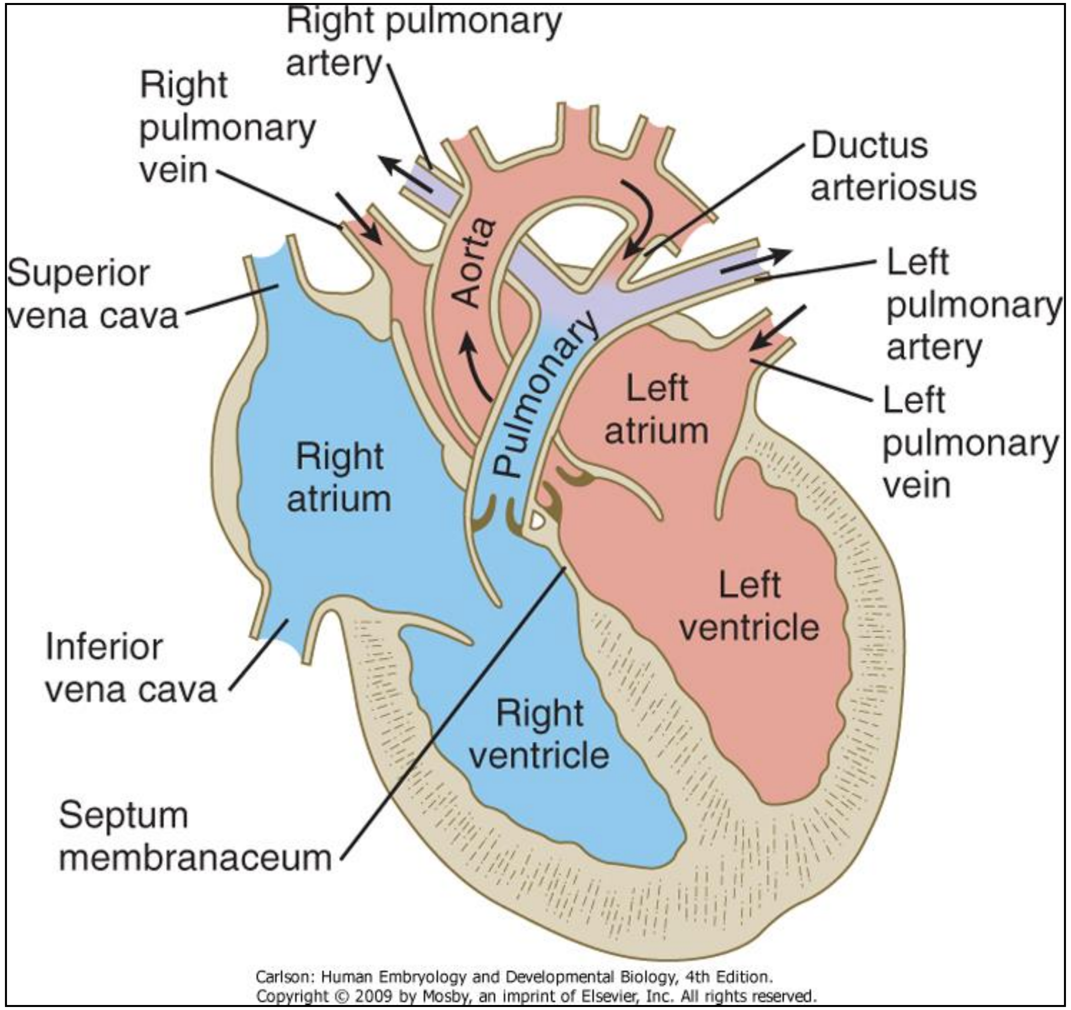
Ductus arteriosus persistens



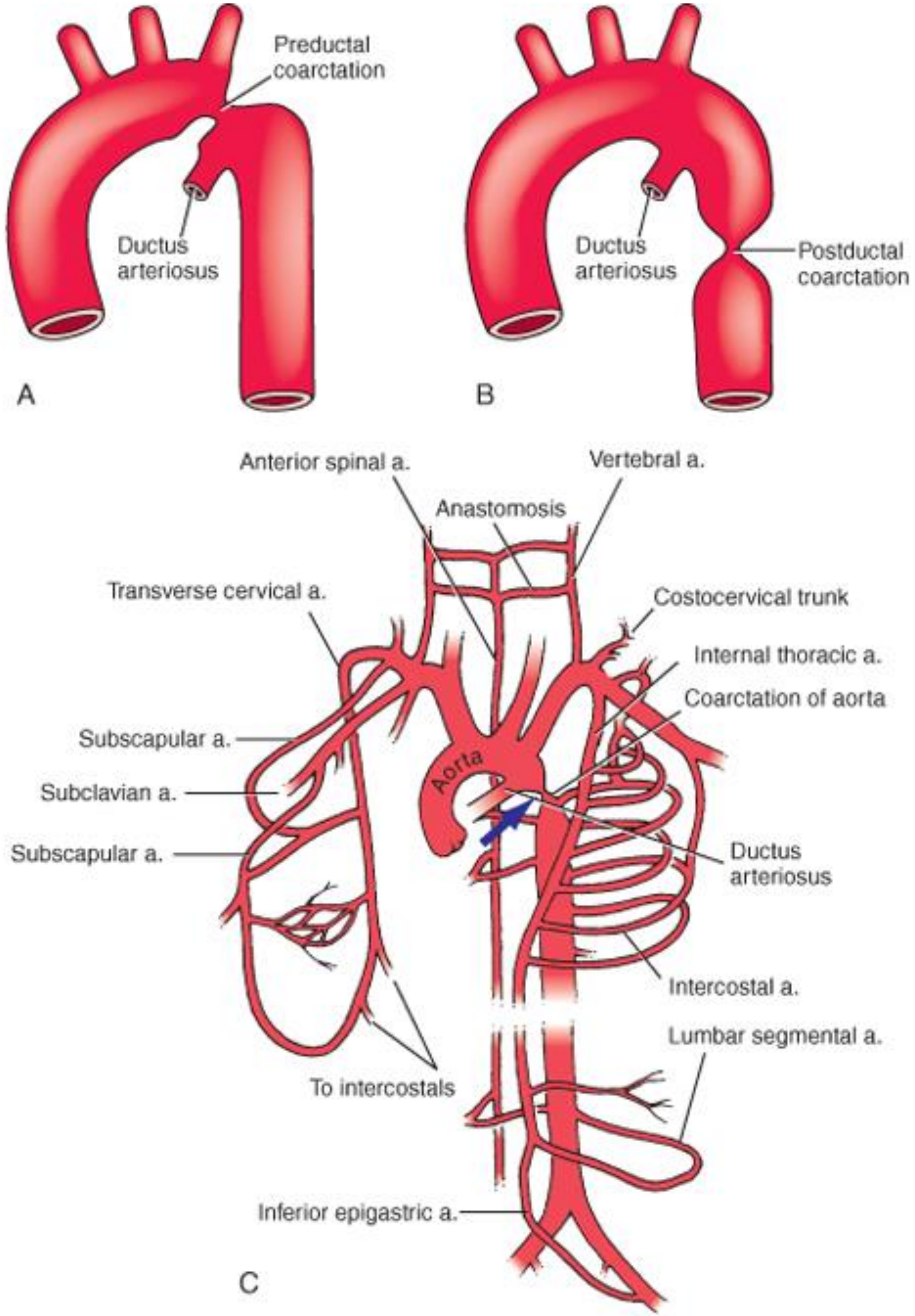
normal



pathological



Patent ductus arteriosus showing the flow of blood from the aorta into the pulmonary circulation. Later in life, pulmonary hypertension may result, causing the reversal of blood flow through the shunt and cyanosis.



Schoenwolf et al: Larsen's Human Embryology, 4th Edition.


THANK
YOU
FOR
YOUR
ATTENTION

References

Lectures of Dr. Ágnes Csáki and Dr. Károly Altdorfer

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